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Laser heating and ablation of carbon–carbon composites

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ABSTRACT

Carbon/carbon (C/C) composites are used as heat shield materials on hypersonic vehicles. Localized ablation can be initiated on C/C composites through targeted heating that results in oxidation and sublimation of the material. The effects of sublimation through targeted heating on C/C composites are studied and compared to graphite by ablating both materials using a 2 kW continuous wave laser in a low-pressure, inert environment. The average temperatures on the back face of both materials under multiple experimental conditions show that graphite experiences a higher average temperature on the back face than C/C composites. Scanning electron microscopy (SEM) micrographs of the ablated C/C composite samples show regions where the fibers ablate faster than the matrix. In addition, fibers exhibit needling and thinning in regions where the matrix ablates before the fibers. The surface of samples outside the ablated region developed a surface coating which was analyzed through X-ray photoelectron spectroscopy and determined to be re-deposition of sublimation gases onto the surface of a sample. The in-depth material response near the ablated region was imaged using X-ray computed tomography (XRCT), and no evidence of internal damage was found. The profiles of the ablated craters obtained using XRCT were fit to a bivariate normal distribution to measure the recession profiles. The results showed that the depth of the craters in the C/C composite was lower than the depth of the crater formed on graphite.

1. Introduction

Carbon/carbon (C/C) composites are used as heat shield materials in high-speed vehicles [1], space capsules [2], solid rocket motors [3–5], aircraft brakes [6,7], and nuclear reactors [8,9]. They are also used for electromagnetic shielding [10–13]. C/C composites relieve the heat load by sacrificially converting solid carbon into gaseous products, a process called ablation [14]. The gaseous products also modify the high-temperature boundary layer that acts as an additional cooling mechanism [15]. Ablation can occur through both chemical and physical processes. At temperatures above 800 K, atomic and molecular oxygen react with carbon to form CO and CO_2. Atomic and molecular nitrogen can also react with carbon to form CN. Furthermore, carbon can also sublime into gaseous products at high temperatures. The sublimation rates of the gaseous products are a strong function of both temperature and pressure. At temperatures above 3000 K, sublimation is the major driving force behind ablation of carbon-based systems [16].

Many studies have been conducted on C/C composite and graphite ablation. However, these studies ablate samples using arc-jets, plasma torches, or high-temperature flames. Under these conditions, carbon samples ablate by both oxidation and sublimation mechanisms. Few studies isolate sublimation mechanisms. Levet et al. have conducted a study on the sublimation of three-dimensional (3D) C/C composites using a uniform heat flux produced from xenon lamps [17]. Geng et al. used a continuous CO_2 laser to compare ablation resistance between C/C and C/C–SiC–Si composites, but did not focus on the underlying mechanisms that dictate the crater morphology [18,19]. This article focuses on the role of sublimation in the ablation of carbon-based composites.

Sublimation of C/C composites is a coupled heat transfer and material-interaction problem, where the microstructure of the material plays a crucial role in the coupling of various processes [20]. C/C composites consist of interwoven weaves impregnated with a carbon matrix. These carbon weaves comprise of bundles of fibers, typically consisting of an ordered structure, undergoing sublimation primarily through localized pit formations [21,22]. These pits appear qualitatively similar to the pits that are observed in highly oriented pyrolytic graphite (HOPG) [23–26]. In contrast, the impregnated carbon matrix could have an ordered graphitized or non-ordered structure [27].

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Regardless of the uncertainty in the relative sublimation rates of the fibers and matrix, differences in the density and structure of carbon fibers and the matrix lead to a phenomenon called differential ablation, in which the matrix and fibers ablate at different rates [28,29]. The differential ablation phenomenon is also affected by the distribution of voids within C/C composites [30]. Furthermore, interweaving of carbon fiber bundles results in anisotropic thermal conductivity for C/C composites [31]. The anisotropic thermal conductivity results in increased heat transport in the in-plane direction along the tows, which reduces heat transport through the thickness of the material. As a result, it has been shown that the ablation of C/C composites is influenced by microstructure (polycrystalline, three-dimensional C/C, fibrous, resin, etc.), the resulting surface area, heat transport, and the inter-coupling of these processes [17,30,32–34].

Although the sacrificial removal of C/C composites is beneficial as a passive cooling approach, high sublimation rates can be detrimental to the survivability of vehicles during high-speed flight. Specific regions of the material can sublime because of targeted localized heating that can form craters within the material. Cratered regions exhibit higher heating, resulting in a cascading effect, where increased heating leads to higher sublimation within the cratered region [35]. Various mechanisms influence the crater morphology. In addition to heating rates, material properties such as carbon weave structure, overall microstructure, and composition of the impregnated matrix will impact the resulting crater profiles.

To understand the interplay of various processes that result in localized sublimation, two-dimensional (2D) C/C composite samples were exposed to a high-power continuous wave laser [36]. To reduce the complexity of gas-phase coupling in ablation, the experiments were conducted in a low-pressure, inert atmosphere to ensure that ablation was driven only by sublimation reactions. The laser focused on depositing energy in a small region of the C/C composite. As energy is absorbed, the temperature of the material increases, resulting in sublimation and the formation of craters. The formation of craters in 2D C/C composites is compared with commercial-grade graphite under different laser power conditions to contrast the similarities and differences between the two materials.

2. Methods

Graphite and C/C composite samples were ablated using a high-power continuous wave laser in a low-pressure, inert atmosphere. The samples were machined into cylinders, and the laser impinged on the circular face of each sample. Temperature measurements were taken in-situ, while the surface response, internal response, surface elemental composition, crater depth, and crater profiles were studied ex-situ using various characterization techniques.

2.1. Materials and preparation

The materials used in this study were graphite and a woven 2D C/C composite. Entegris EDM-3 POCO graphite was selected as it is well-characterized by the manufacturer, and the internal material structure is composed of ultra-fine particles [37]. A 2D woven material manufactured by Americarb was chosen as the C/C composite [38]. The commercial C/C composite from Americarb is CC-16HCMX. All graphite and C/C composite samples were cut from a flat plate using a hole saw attached to a drill press. The graphite and C/C composite samples were cylinders with diameters and heights of 0.875 and 0.5 inches, respectively. An image of a C/C composite sample used during the experiment is shown in Fig. 1a, and an image of a graphite sample is shown in Fig. 1b.

The microstructures of the C/C composite and graphite samples are shown in Figs. 1c and 1d, respectively. The C/C composite is formed by attaching multiple carbon fibers with a carbon-based pitch to create bundles of fibers. The bundles are then linked in a woven pattern, and the entire structure is attached using the same carbon-based pitch [21,22]. The C/C composite has an average volume fraction of 0.94, and the average tow thickness is 2.41 mm. The manufacturing process of C/C composites results in void spaces throughout the material, as shown in Fig. 1c. The combination of void spaces and varying orientations of fibers results in the material having a non-uniform internal structure. The graphite used in the experiment comprises ultra-fine graphitic carbon particles with an average size below 5 μm, resulting in a uniform internal structure [37].

Table 1
Parameters of the laser and the test times used for the three conditions in the experiments.

Condition number	Laser power [W]	Beam width [mm]	Time interval [s]
1	900	6	150
2	1350	6	150
3	1800	6	25–40

Table 2
Laser heat flux parameters.

Condition number	Peak heat flux [W/cm^2]	Average heat flux [W/cm^2]
1	5390	267.36
2	8086	401.05
3	10 780	534.73

2.2. Experimental setup

A schematic of the experimental setup is shown in Fig. 2, and an image of the setup is shown in Fig. 3. The continuous wave IPG YLR laser used for the experiments had a maximum power of 2 kW and a wavelength of 1070 nm. The laser beam had a near-Gaussian profile with an M^2 value of 1.1, according to the manufacturer. The Gaussian profile had a full width half maximum (FWHM) of 3.93 mm along the major axis and 3.42 mm along the minor axis. The power of the laser was set manually by the operator using software that controlled the laser output.

The experimental conditions are shown in Table 1. A low power condition of 900 W, a medium power condition of 1350 W, and a high power condition of 1800 W were selected for the experiments. Multiple samples were exposed to the same laser condition. Three graphite and three C/C composite samples were ablated under low power conditions. Two graphite and three C/C composite samples were ablated under medium power conditions. Two graphite and two C/C composite samples were ablated under high power conditions. The complete list of experimental samples with their corresponding sample number and ablation conditions is included in Table S1 in the Supporting Information. These conditions were chosen to include varying degrees of sample damage. The test times were selected at 150 s so that the samples could approach steady-state temperature conditions. Under the high power condition of 1800 W, the back face temperature of the sample would exceed the dynamic range of the infrared camera (2000 °C) before an elapsed time of 150 s. To obtain accurate temperature data throughout the entire experiment, the laser was turned off once the maximum temperature on the back face reached 2000 °C. The total experiment time for samples in condition 3 ranged from 25 to 40 s.

The heat flux generated by the laser is a normal distribution, mainly dependent on the beam width and the laser power [36]. An estimate of the peak and average heat fluxes for each condition is presented in Table 2. A full estimate of the heat flux over the length of the sample is shown in Fig. S1 in the Supporting Information.

The laser was directed at the sample through a window in the vacuum chamber. A series of images that describe the state of a C/C composite sample throughout an experiment are shown in Fig. 4. The process is representative of each of the tested samples. Inside the

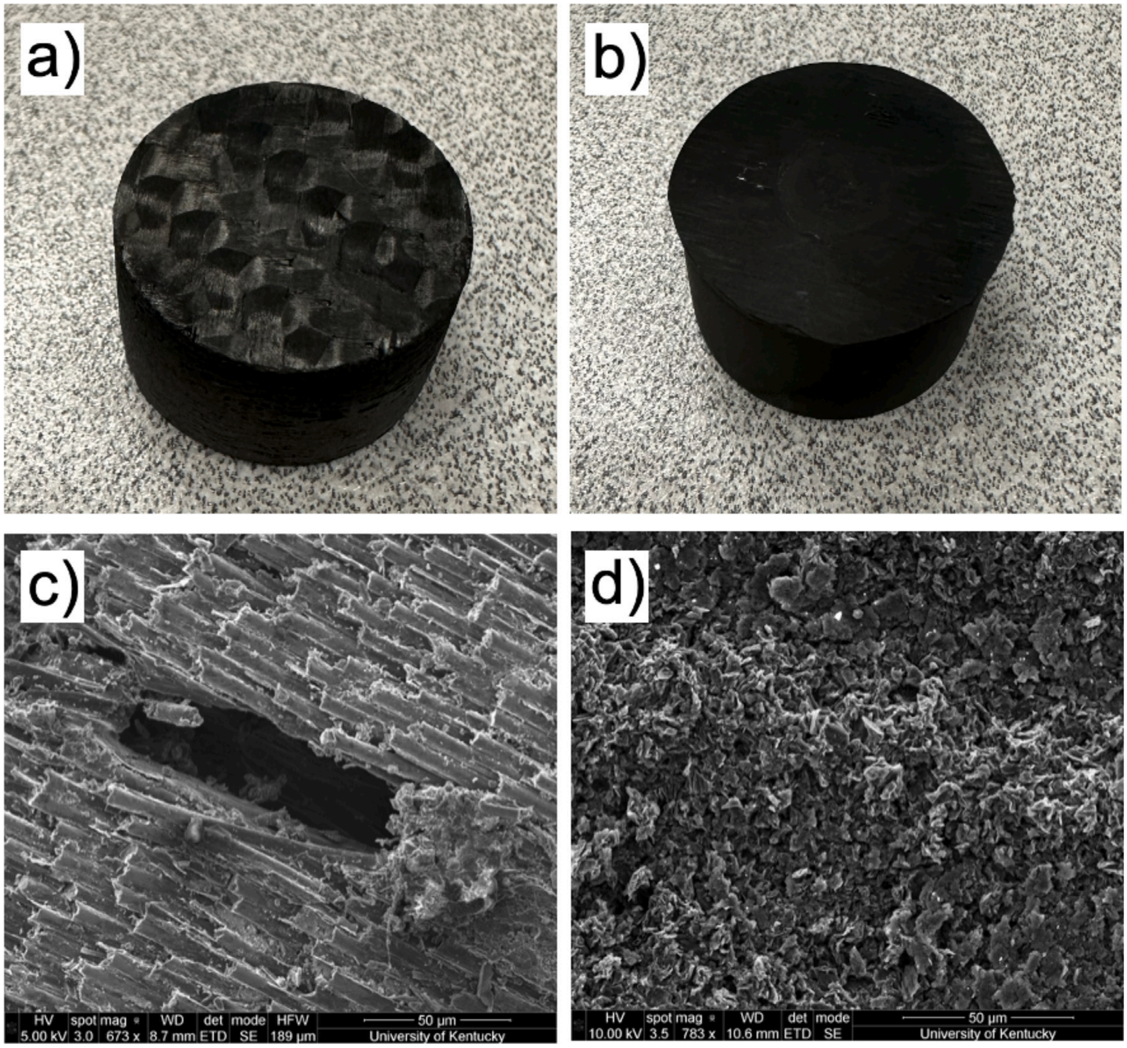

Fig. 1. Images of non-ablated samples and SEM micrographs of non-ablated microstructures. (a) C/C composite sample, (b) graphite sample, (c) C/C composite microstructure depicting the fibers, matrix, and void regions, (d) graphite microstructure depicting the ultra-fine graphitic carbon particles.

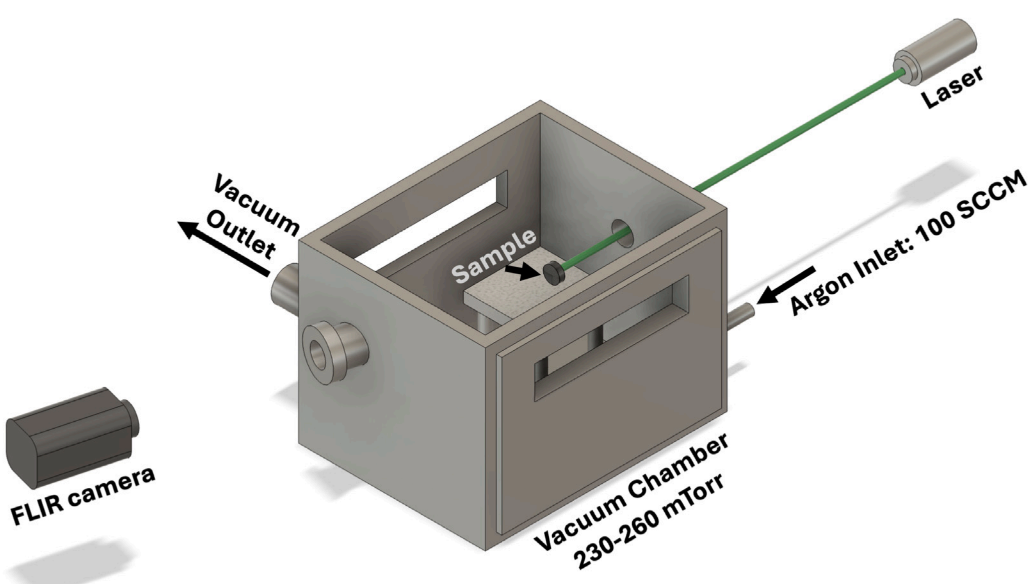

Fig. 2. Schematic of experimental setup. The sample sits on an aluminum oxide table inside the vacuum chamber. The laser is aimed at the sample through a viewing window in the front of the chamber, whereas the FLIR camera observes the sample through an angled viewing window in the back of the chamber.

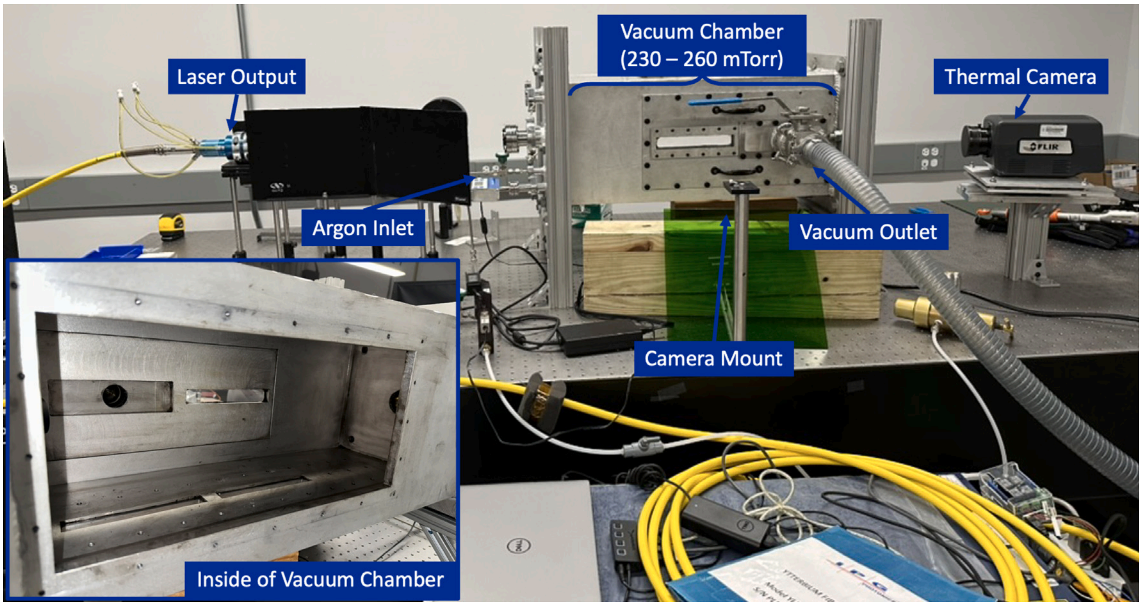

Fig. 3. Photo of experimental setup. Important components have been labeled.

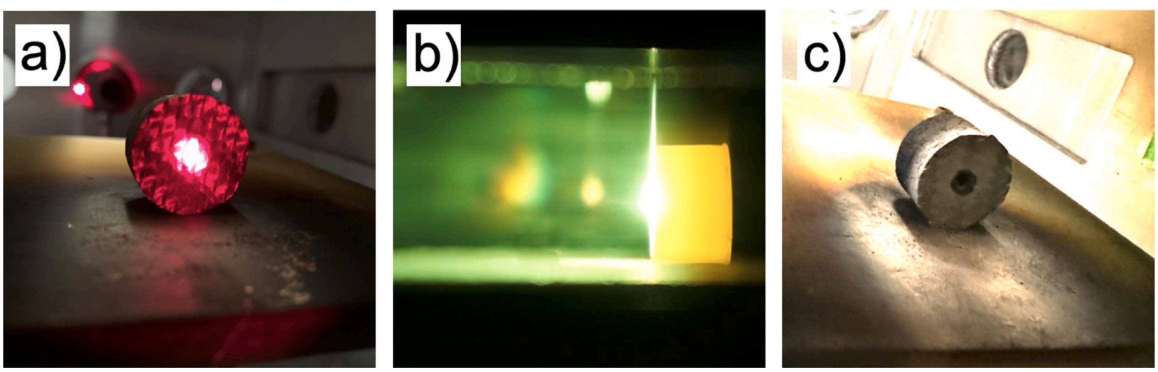

Fig. 4. Images of a C/C composite throughout an experiment, (a) before, (b) during, and (c) after. The red dot in (a) is the guide beam used to align the sample with the laser.

vacuum chamber, samples were manually placed on an aluminum oxide plate and oriented to point a circular face towards the laser. The samples were placed at a constant distance from the edge of the plate using an alignment tool. A guide beam was used to orient the sample so the laser would impinge on the center, as shown in Fig. 4a. After the sample was in place, the vacuum chamber was brought to 230–260 mTorr. The vacuum outlet was behind the sample on the side of the vacuum chamber, as shown in Figs. 2 and 3. Argon entered the chamber at approximately 100 SCCM. The viewing windows were located on the sides of the chamber for observation. An image of a C/C composite sample during an experiment, observed through a viewing window, is shown in Fig. 4b. After the time for a given condition elapsed, the laser was powered off. The sample was then allowed to cool for a period, preventing post-test oxidation and nitridation. After the sample cooled, air was slowly introduced into the chamber until atmospheric conditions were reached. The sample was then removed and replaced. An ablated sample before removal is shown in Fig. 4c.

A Forward Looking InfraRed (FLIR) A8581 camera was pointed at the back face of the sample and mounted to the optical table. The camera observed the sample through an angled window in the back of the vacuum chamber, as shown in Fig. 3. The FLIR camera was used to acquire temperature data. Two examples of temperature data are shown in Fig. 5a for graphite, and Fig. 5b for the 2D C/C composite. These images were captured after 140 s under condition 1 with a laser power of 900 W. The dynamic range of the FLIR camera during the experiment was between 800 °C and 2000 °C. A graphite sample with a thermocouple in an OTF-1200X-S clamshell furnace was used to calibrate the FLIR camera [36]. The temperature correction equation was developed by comparing the thermocouple and FLIR data. Further details regarding the calibration process are provided in Sec. S3 of the Supporting Information [36].

$$T_{corr} = T(1.01 + 4.67 \times 10^{-5}\, T)$$

Data were recorded from the start of the experiment until the sample cooled below 800 °C, the minimum temperature the FLIR camera could record. The camera recorded at 10 frames per second with a resolution of 1024 × 1024 pixels. The back face temperature of the sample was recorded at each pixel for every frame throughout the recording duration. The data were exported as a collection of comma-separated value (CSV) files and used to extract information about the temperatures on the back face of the sample throughout the experiment.

2.3. Post-test characterization

2.3.1. Scanning electron microscopy

Scanning electron microscopy (SEM) was performed using a Quanta FEG 250 E-SEM scanning electron microscope to study the surface

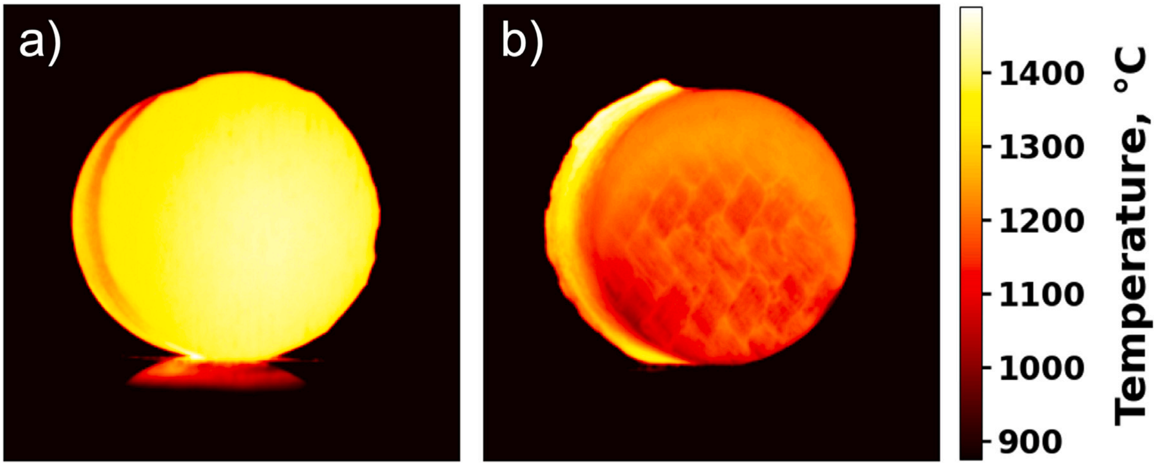

Fig. 5. Two frames of FLIR camera data for samples tested under condition 1. (a) Graphite and (b) C/C composite, after 140 s. Condition 1 exposes the samples to a laser beam with a power of 900 W.

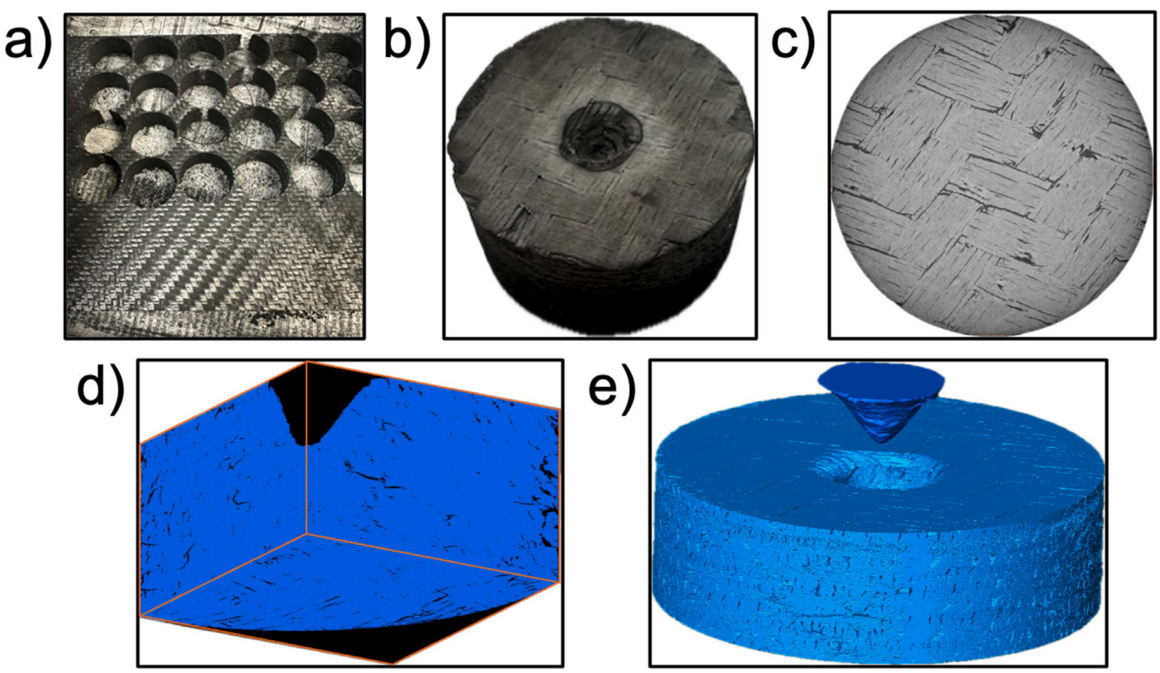

Fig. 6. Overview of the XRCT workflow for C/C composite. (a) C/C composite plate from which samples were machined, (b) C/C composite sample, (c) tomograph of C/C composite from XRCT, (d) labeled material of C/C composite obtained from universal threshold, (e) segmented C/C composite.

response and non-ablated surface of the materials. The voltage in kilovolts (kV), the spot size in nanometers (nm), the working distance in millimeters (mm), and the detector information can be found at the bottom of each SEM image.

2.3.2. X-ray computed tomography

X-ray computed tomography (XRCT) is a non-invasive method of capturing the internal structure of a sample. Using the data produced by the XRCT scan, a three-dimensional digital representation of the sample and crater profile is obtained. The XRCT machine used was a HeliScan Mk-2 MicroCT from ThermoFisher Scientific. The voltage, current, exposure time, and voxel size for each scan are listed in Table S3 in the Supporting Information. For each scan, 1800 projections over a 360 ° rotation were collected. A propagation-based phase-contrast reconstruction method was used to convert projections into tagged image file format (TIFF) tomographs [39]. Multiple tomographs representing single-voxel-thick cross-sectional images of the sample were produced and exported as TIFF stacks.

Fig. 6 shows the XRCT workflow for a single sample. Fig. 6a shows the C/C composite plate from which the samples were machined. Fig. 6b shows an ablated C/C composite sample. A single TIFF stack from a C/C composite scan is shown in Fig. 6c. The TIFF stacks of the tomographs were imported into Avizo-Amira software, developed by ThermoFisher Scientific, to extract the three-dimensional crater profile from the sample. First, the tomographs were filtered to remove unnecessary noise. Then, the voxels representing the crater were manually selected for a slice, and the crater voxels for a slice further down the stack were selected. All voxels between the two manually selected layers were added to the selection using interpolation. After selecting voxels for all slices, the interpolated sections were verified and adjusted if necessary. The selected voxels representing material rather than grayscale values are added to a new binarized TIFF stack. Three slices of the binarized TIFF stack in the *XY*, *XZ*, and *YZ* planes are shown in

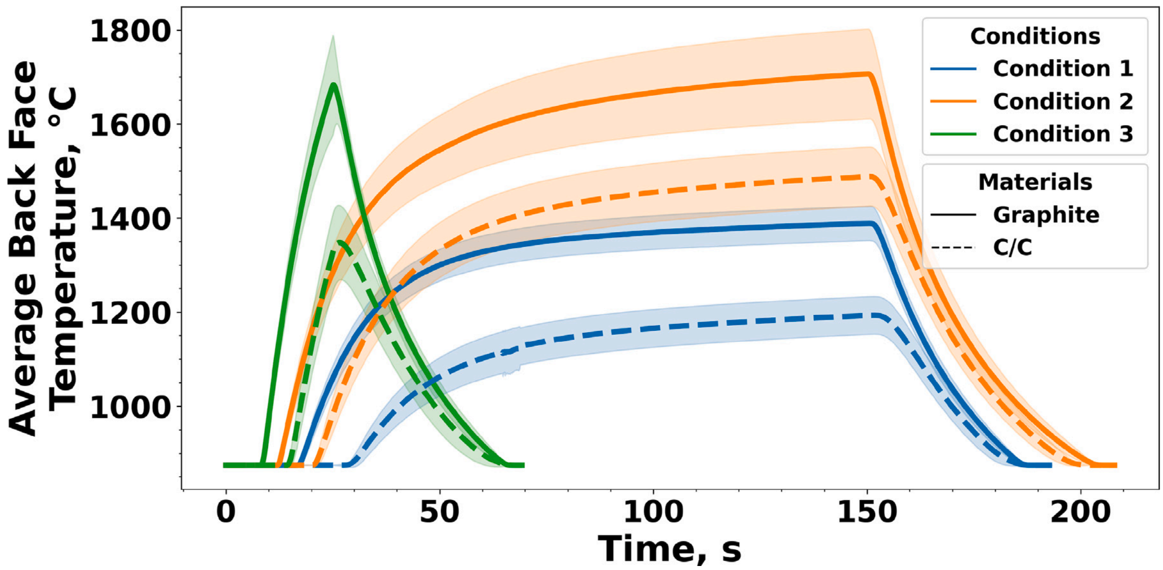

Fig. 7. Average of mean back face temperature between all samples for all conditions. The shaded region represents the average standard deviation of the back face temperature for the samples within the same condition. Figures S2–S7 in the Supporting Information show the individual mean temperature of the back face for all samples.

Fig. 6d. Finally, a three-dimensional surface object, shown in Fig. 6e, was generated from the binarized material voxels.

2.3.3. X-ray photoelectron spectroscopy

X-ray photoelectron spectroscopy (XPS) was performed using a ThermoFisher Scientific K-alpha X-ray photoelectron spectrometer using an Al-Ka low power monochromator. The experiment was performed automatically using the Avantage software, developed by ThermoFisher Scientific, with manual input parameters. The K-alpha spectrometer is equipped with a flood gun, which uses low-energy ion beams and low-energy electrons to prevent issues with differential charging. The flood gun feature was used throughout the scans. X-ray photoelectron spectroscopy was performed on one graphite and one C/C composite sample. For both samples, three scanning locations were selected for redundancy. Each location had a spot size of 300 μ m. A survey scan and a high-resolution carbon scan were conducted at each location. The survey scan measures over a wide range of binding energies (0 eV to 1350 eV) to determine the elements that make up the surface composition of the sample. The survey scan had a step size of 1 eV, a pass energy of 200 eV, and a dwell time of 10 ms. The high-resolution carbon scan measures a small range of binding energies (279 to 299 eV) to isolate carbon. The carbon scan gives information on the elemental state of carbon on the surface of the sample. The carbon scan had a step size of 0.1 eV, a pass energy of 50 eV, and a dwell time of 50 ms.

3. Results and analysis

The following section presents the results and analysis of the experiments. The temperature data were analyzed to compare the thermal response of both materials. The in-depth and surface material response to the laser is shown using XRCT and SEM images and compared to non-ablated material regions. The surface of both materials was analyzed using XPS to determine the elemental composition. The depth of each crater is provided, as well as an analysis of the crater profiles.

3.1. Thermal response

The average of the mean back face temperature for all samples under each condition is shown in Fig. 7 as extracted from the FLIR camera data. A border for the back face was defined for each sample, and the mean temperature was computed by averaging the temperature of every pixel within the back face border. In Fig. 7, the samples are grouped by condition and material, and the average of the mean back face temperature for each group is shown. For instance, three C/C samples were exposed to condition 1. The mean temperature of the back face of each C/C sample under condition 1 was extracted from the FLIR data, and Fig. 7 shows the average of the mean back face temperature of those samples. Figures S2–S7 in the Supporting Information show the average back face temperature for all samples. In addition, the shaded region represents the average standard deviation within each sample. The standard deviation for each sample is computed from the FLIR camera data between the pixels within the back face, similar to the average temperature, and the average standard deviation for each group is shown. For the third condition, the average temperature of the back face is provided for a single sample for each material, as the samples were exposed at different times.

The dynamic range of the FLIR camera was between 800 °C and 2000 °C, which omits the initial temperature readings of the experiments shown in Fig. 7. The lack of initial readings appears to be a delay between the different experiments. However, the delay can be attributed to the higher heat fluxes generated during the high power conditions, causing the back face temperature to rise faster. The initial time of the experiments, although not visible in Fig. 7, is when the laser is turned on and hits the surface of the sample.

It can be seen that all graphite samples exhibit a higher average back face temperature than all C/C composite samples for each condition. However, presenting the average temperature of the back face for the samples omits the variation in temperature of the back face. To demonstrate the variation, Fig. S8 is shown in the Supporting Information, containing the average coefficient of variation (CV) between the samples within the same condition. As also shown in Fig. 5, the weaves in the C/C composite create a significant variation in temperature on its back face. In general, it is observed that the variation is larger for the C/C composite samples than for the graphite samples.

Although the graphite sample appears to be more uniform, it still has a non-negligible coefficient of variation. The coefficient of variation in graphite arises from the diffusion of heat on the back face of the material. Graphite is an isotropic material, with isotropic thermal properties. Therefore, the heat transfers to the back face quicker in

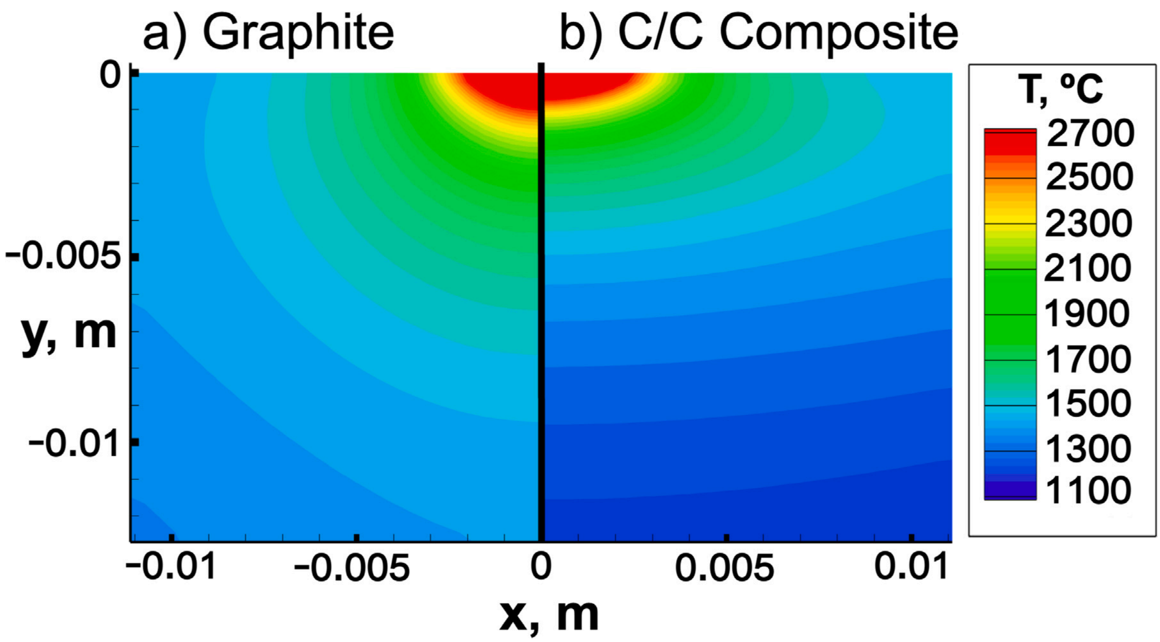

Fig. 8. Material response simulation of both materials under condition 1 after 120 s. (a) Graphite and (b) C/C composite.

graphite compared to the C/C composite. Once the heat reaches the back face, it expands radially, causing the observed temperature variation within the back face of graphite. In contrast, the C/C composite is an anisotropic material with a lower through-plane conductivity. Therefore, the heat expands radially in the C/C composite before reaching the back face. However, the presence of fibers, matrix, and voids within the C/C composite is a more prominent factor in creating variations in the temperature of the back face than the differences in thermal diffusion within the materials.

To understand the effect of thermal diffusion further, material response simulations were performed for each material under the three conditions. The simulations were performed using the material response module of the Kentucky aerodynamic and thermal-response system (KATS-MR) [40]. The simulated geometry was an axisymmetric slice of the samples, with a symmetric boundary condition applied to the left, front, and back faces to establish the axisymmetric configuration. The top face was exposed to the peak heat flux (see Table 2). The remaining faces, along with the top face, used a re-radiation boundary condition. The recession was disabled to focus on the temperature distribution. The temperature distribution in the samples after 120 s for condition 1 is shown in Fig. 8. Furthermore, the same simulations were performed for conditions 2 and 3, and the results are shown in Figs. S9 and S10 of the Supporting Information. The temperature contour of each material is placed side-by-side for comparison.

The larger in-plane conductivity of the C/C composite causes the heat to concentrate within the front of the composite and diffuse within the plane. The concentration of the heat within the front of the C/C composite results in lower temperatures in the back face when compared to the graphite material, as observed in Fig. 7. Furthermore, diffusing the heat within the front face also results in larger cooling via radiation from the exposed front and side faces, further reducing the average back face temperature of the C/C composite compared to the graphite material. The isotropic conductivity of the graphite material allows the heat to traverse the through-plane direction faster than it does in the C/C composite samples.

To emphasize the effect of conductivity differences on the lower back face temperature of the C/C composite, a second set of material response simulations was performed by turning on sublimation. The sublimation rates are obtained from Refs. [41–43]. The simulations model the experiments under condition 1. Further details can be found in Ref. [44]. The ablation enthalpy and emissivities are equal for both graphite and the C/C composite. The temperature contour for both materials after 120 s is shown in Fig. 9. Under these conditions, and with ablation enabled, the graphite material still shows a higher back face temperature than the C/C composite. The simulations shown in Fig. 8 present an average back face temperature of around 1100 °C for the C/C composite, and approximately 1400 °C for the graphite material. As ablation is enabled, as shown in Fig. 9, the average back face temperature for the C/C composite increases to around 1200 °C, whereas it increases to roughly 1500 °C for the graphite material. Considering that the conductivity is the main difference in material properties for each material, aside from a 20% difference in density, we conclude that the higher in-plane conductivity exhibited by the C/C material, in comparison to the graphite material, is the main cause for the lower back face temperature in C/C composites.

3.2. Material response

3.2.1. Evaluation of microstructure response

The surface material response and ablated microstructure of graphite and C/C composite samples are shown in Fig. 10. The crater formed on the C/C composite sample under condition 1 is shown in Fig. 10a. Compared to the graphite crater under the same condition (Fig. 10b), the influence of the heterogeneous material architecture of C/C composites on damage is apparent. Figs. 10c and 10d show the surface of each material adjacent to the crater for C/C composite and graphite, respectively. A surface coating has developed on both materials outside the crater region. Samples under conditions 2 and 3 were also imaged and showed similar features. The surface coating is discussed in the following section, Section 3.2.2.

The graphite crater in Fig. 10b appears smooth and circular, whereas the C/C composite crater appears disorganized and ellipsoidal. Voids and orientations of fibers, discussed in Section 2.1, that lead to anisotropic heat conduction throughout the C/C composite are some of the reasons for the cause of disordered ablation [17,32]. In contrast, heat conduction in graphite is more isotropic because of the uniform internal structure of the material. Additionally, the ultra-fine particles that comprise the internal structure ablate entirely once sublimation temperatures are reached. Evidence is shown by comparing the ablated graphite microstructure shown in Fig. 11a with the non-ablated

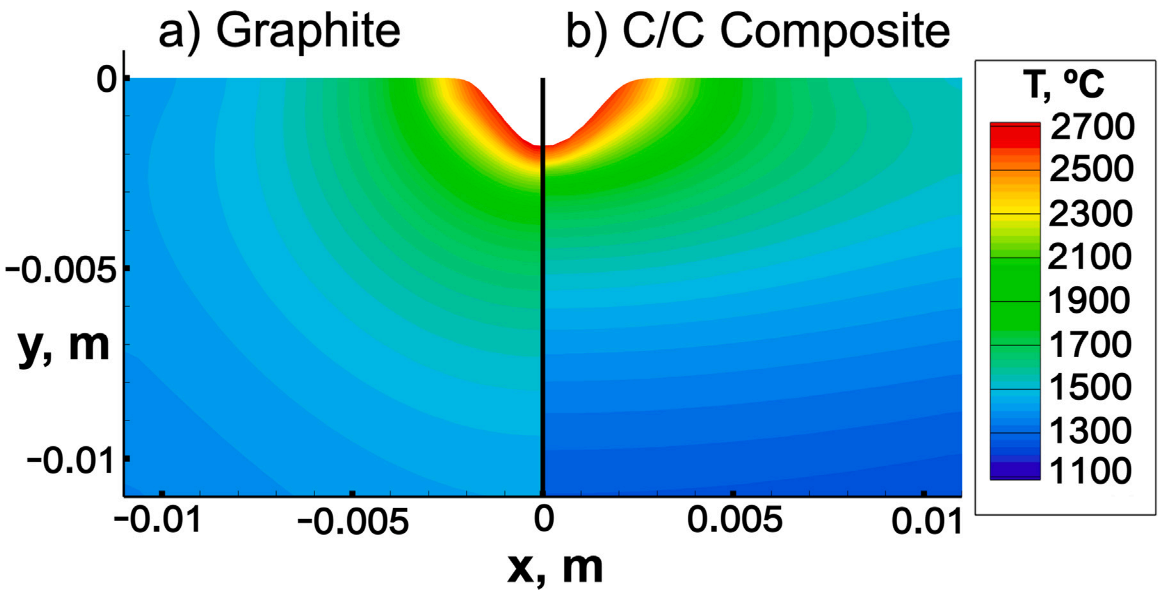

Fig. 9. Material response simulation of both materials under condition 1 after 120 s with ablation enabled. (a) Graphite and (b) C/C composite.

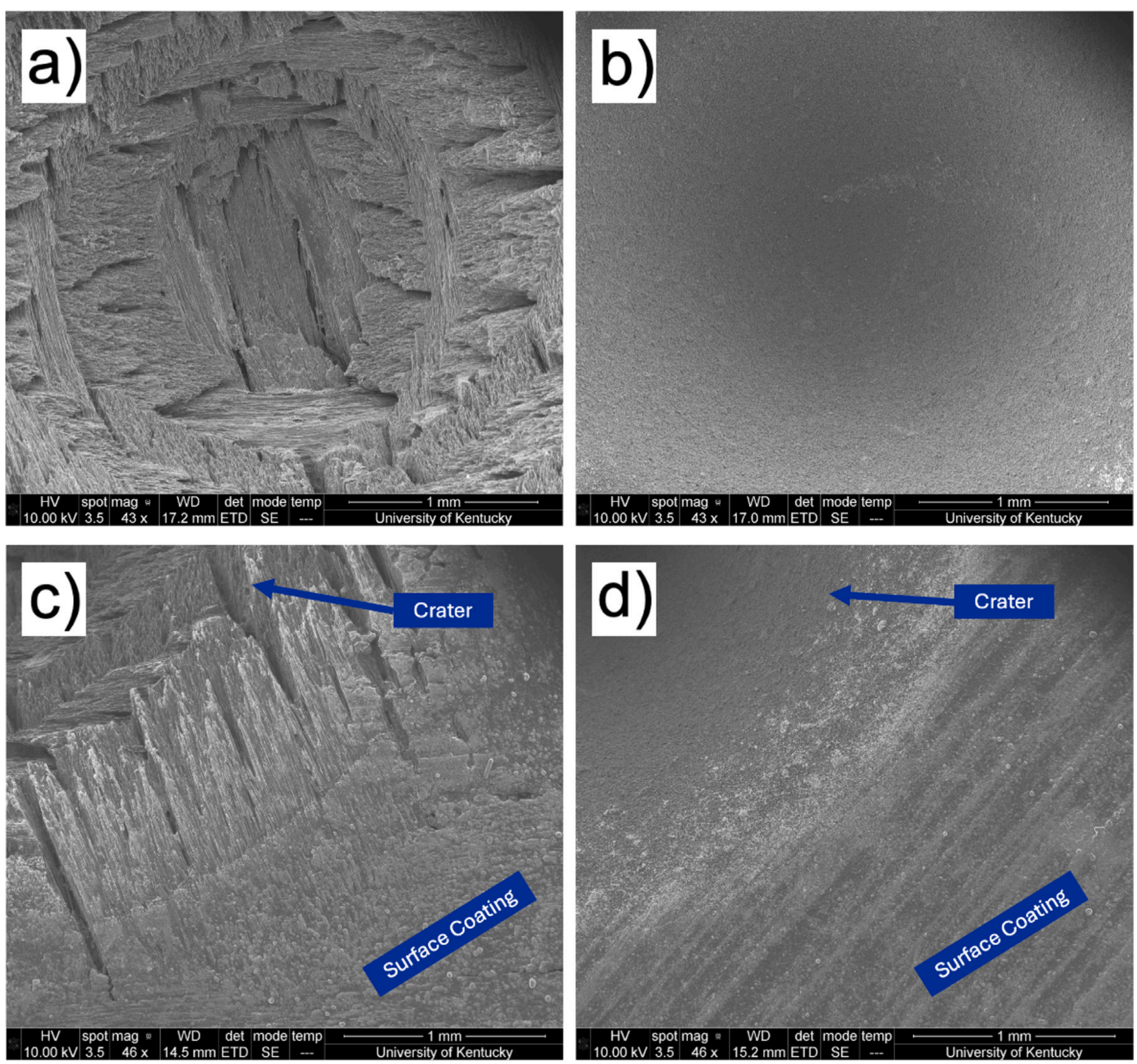

Fig. 10. SEM micrographs of the surface of an ablated graphite and C/C composite. (a) C/C composite crater, (b) graphite crater, (c) surrounding region of C/C composite crater, (d) surrounding region of graphite crater.

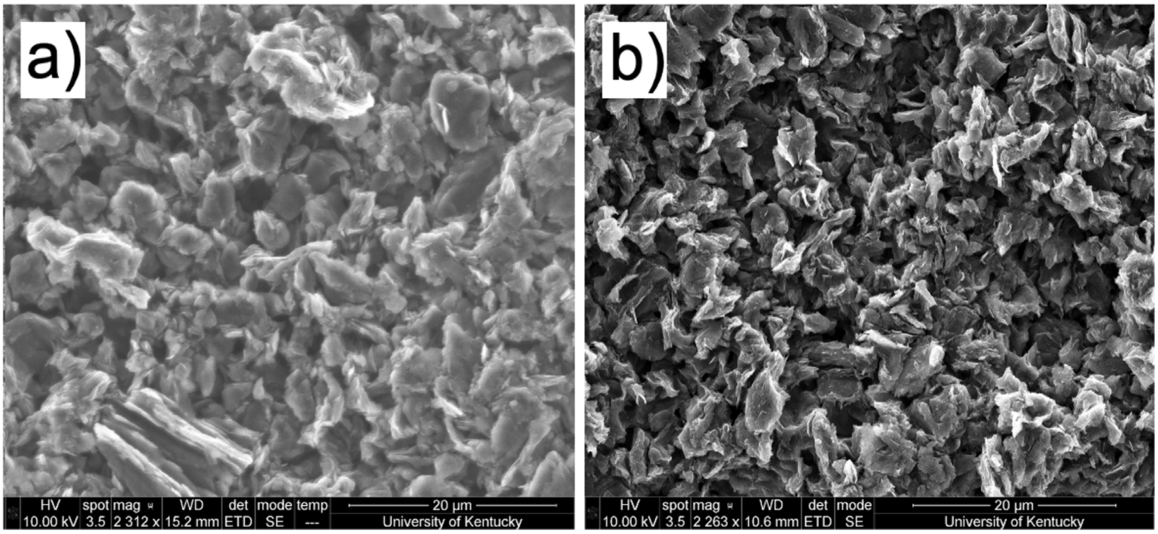

Fig. 11. SEM images of ablated graphite and non-ablated graphite microstructure. (a) Ablated graphite microstructure and (b) non-ablated graphite microstructure.

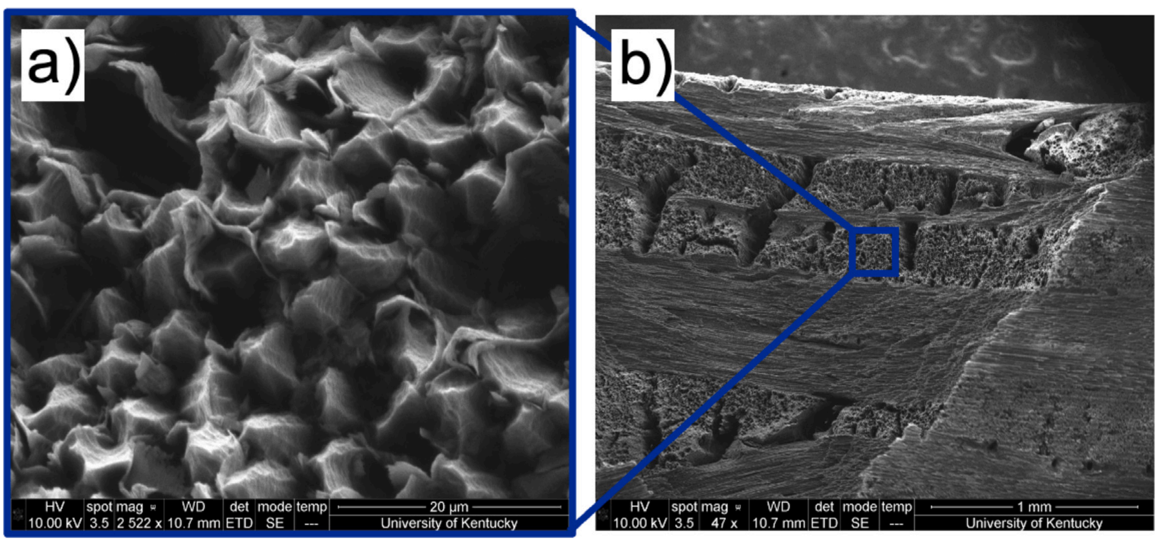

Fig. 12. SEM micrographs of C/C composite ablated microstructure showing the differential ablation phenomenon. (a) Fibers receding into a thin wall of surrounding matrix, (b) location of the preceding figure on the crater.

graphite microstructure shown in Fig. 11b. There is no difference between the ablated and non-ablated graphite microstructure. These ablation characteristics of the graphite samples explain the uniform crater shapes seen in Fig. 10b.

Conversely, C/C composites have a non-uniform internal structure and three distinct material phases: bundles of fibers or weaves, carbon matrix, and voids. The densities and structure of the fibers and matrix are not the same, leading to two different ablation rates in a phenomenon known as differential ablation [28,29]. Differing ablation rates between the fibers and matrix have been frequently observed under oxidation or a combination of oxidation and sublimation. In these cases, the matrix ablates faster than the fibers, and the fibers form smooth, conical, needle-like shapes [32,45,46]. However, previous work has shown that fibers tend to ablate at a rate equal to or slightly faster than that of the matrix under sublimation [17,46–48]. Levet et al. have simulated sublimation behavior of a 3D C/C composite using a reactivity ratio of 5:1 for fibers and matrix, respectively [17]. Fig. 12a shows receded fibers surrounded by a thin wall of matrix, which supports observations made in previous studies [17,46]. The fibers have blunt ends with faceted sides in contrast to the smooth, conical, needle-like fibers seen in oxidation cases [32,45,46]. The observed shape of fibers supports the results seen in sublimation cases [17,46]. The location of Fig. 12a on the crater is shown in Fig. 12b. However, a larger cluster of fibers with a faceted, needle-like shape is also observed and shown in Fig. 13a. While the matrix is still shown surrounding some fibers, there are regions where the matrix cannot be seen. The location of Fig. 13a on the crater is shown in Fig. 13b. The lack of matrix compared to similar studies can partly be attributed to differences between matrix distribution in 2D and 3D C/C composites. The lack of matrix affects the heat conduction throughout the bundles of fibers, explaining why ablation is seen further down the length of the fibers.

In these previous studies, sublimation experiments have been conducted using 3D C/C composites [17,46]. The 3D C/C composites used in these studies were formed from unidirectional bundles of fibers woven into a 3D orthogonal pattern. The process results in empty spaces throughout the material filled with matrix [17,46]. In contrast, 2D C/C composites contain orthogonal woven bundles of fibers affixed with matrix. The contrasting manufacturing processes between the 2D and 3D C/C composites result in varying amounts and distributions

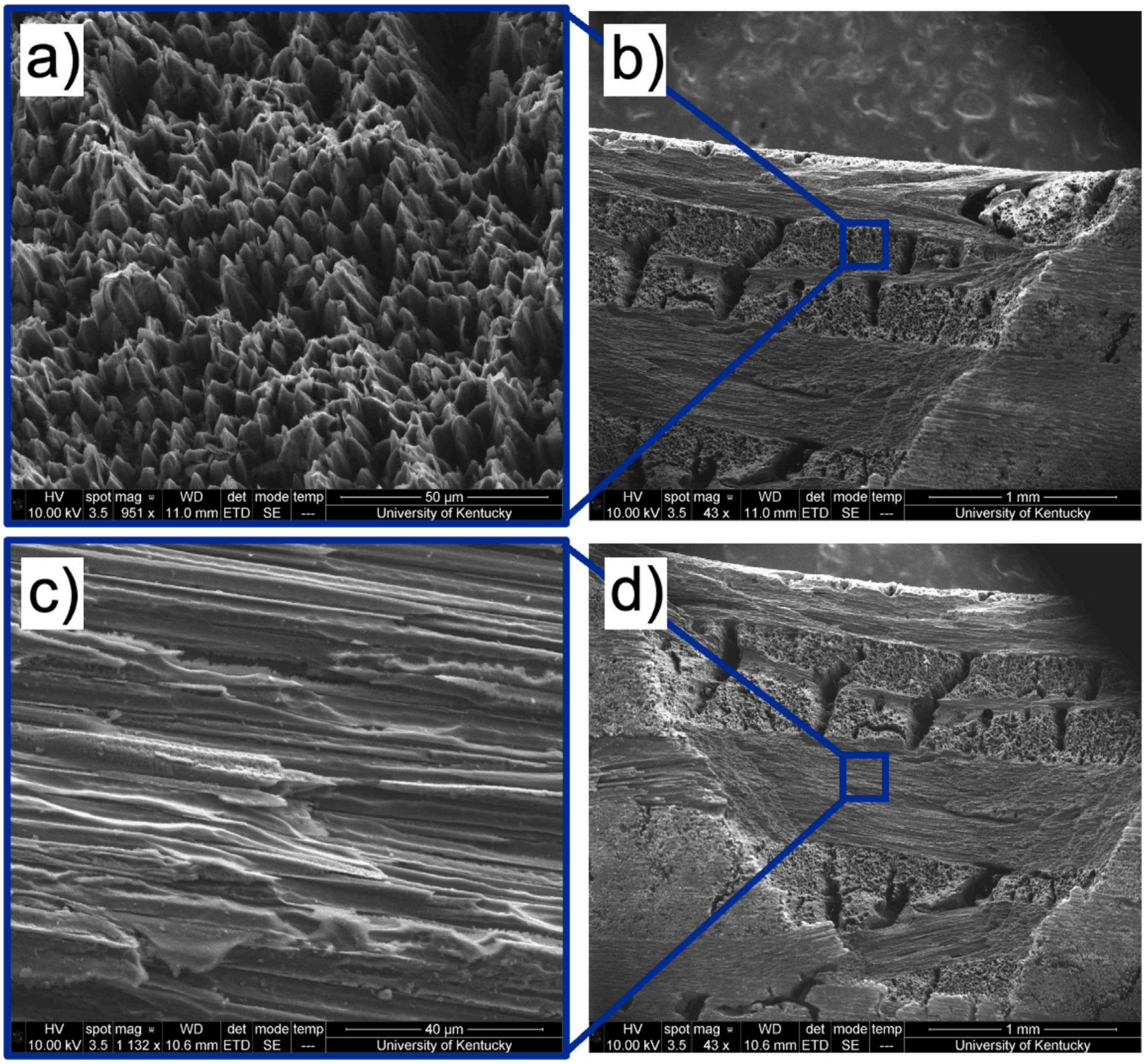

Fig. 13. SEM micrographs of C/C composite ablated microstructure showing needling and thinning of fibers. (a) Cluster of fibers showing needling in the absence of the surrounding matrix, (b) location of the preceding figure on the crater, (c) thinning of fibers, (d) location of the preceding figure on the crater. The sample was cut in half along the crater to image the exposed ends of fibers in the tow direction.

of matrix throughout the material. These differences would affect the localized sublimation of fibers and matrix. Furthermore, a 3D C/C composite would contain fibers oriented in a direction only normal to the applied heat flux. However, for 2D C/C composites investigated in this study, the bundles of fibers were both normal and tangential to the applied heat flux by the laser.

Additionally, previous studies describing the ablated microstructure under sublimation of C/C composites have utilized different heating methods. Levet et al. used xenon lamps to produce a uniform heat flux on the surface of a sample [17]. Vignoles et al. have performed a study using an arc image furnace in an argon atmosphere to sublime samples [46]. In both experiments, the entire face of a sample is exposed to the heat flux. Here, the Gaussian shape of the heat flux generated by the laser focuses the heat load within a 6 mm spot on the surface of the sample, leading to the formation of a crater. The Gaussian-like decay of the laser beam profile used in this study also contributes to the differences in the observed microstructure features from those seen in the aforementioned studies.

Fibers orthogonal to those previously discussed are shown in Fig. 13c. These fibers exhibit thinning, where the diameter of the fibers decreases as the fibers ablate. Reductions in diameter are not along the entire length of the fibers but are localized. Additionally, diameter reductions are biased towards the side of the fibers that are not exposed to the heat flux. The thinning of fibers has been observed in C/C composite ablation under oxidation [46,49]. However, thinning similar to what is shown in Fig. 13c has not been reported under sublimation. The orientation of fibers and the Gaussian laser profile affect the localized ablation of the C/C composite microstructure, and contribute to the thinning shown in Fig. 13c. The location of Fig. 13c on the crater is shown in Fig. 13d.

Levet et al. claim that the faster ablation rate of the fibers, as shown in Fig. 12a, is caused by the matrix being highly graphitized and denser than the fibers [17]. If this were the case, fibers should always ablate faster regardless of the ablation mechanism, which is not true under oxidation. We believe that the inverse differential ablation effect observed in sublimation is related to differences in the thermal stability of the matrix and the fibers. Carbon fibers have crystalline structures with graphitic layers [50–52]. The matrix of the composite investigated here is formed through polymer infiltration-pyrolysis, which results in glassy carbon [53–55]. Glassy carbon is thermally more stable than its crystalline graphitic counterpart [56–58]. Sublimation is a phase-change process that has a non-linear relationship with temperature. The difference in thermal stability could explain why fibers ablate faster than the matrix, as shown in Fig. 12a. However, we also see

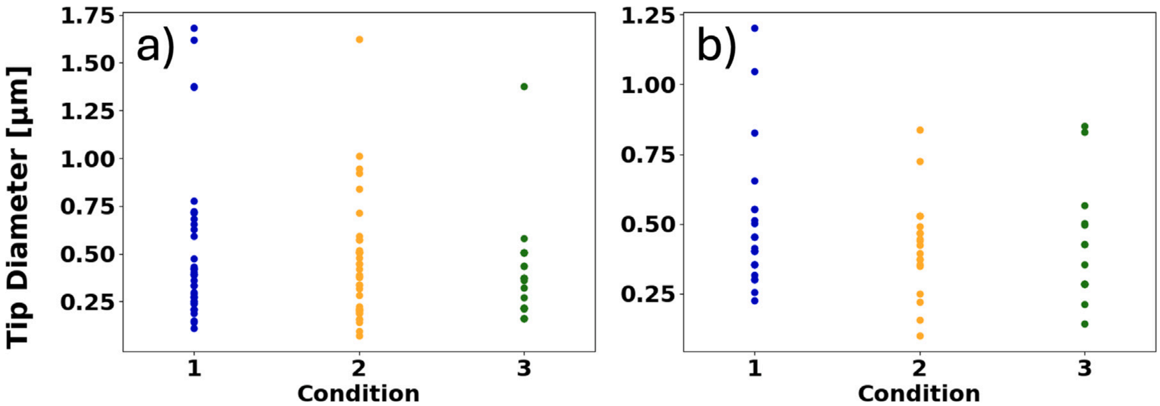

Fig. 14. Tip diameter measurements from each condition in (a) locations near the top of the crater and (b) varying locations within the crater.

regions with less matrix compared to others. There are two possible reasons for this observation. First, the region shown in Fig. 13a could lack the matrix material. However, it is also possible that the matrix has completely ablated away, leading to the conventional differential ablation scenario, where the matrix ablates faster than the fibers. This conventional differential ablation scenario would still exist in sublimation of 2D C/C composites because the overall volume fraction of carbon matrix is lower compared to 3D C/C composites.

The tip diameter of fibers across each condition were measured to determine if there was a correlation between heat flux and final fiber shape. Tip diameter measurements were obtained from SEM images using a Python script. Measurements near the top of the crater are shown in Fig. 14a, and measurements at varying locations within the crater are shown in Fig. 14b. The SEM images used to obtain these measurements are shown in Figs. S11–S14 in the Supporting Information. Two points on each end of the tip diameter are selected, and the distance between these two points in pixels is calculated. The tip diameter can then be obtained from the ratio between pixel size and length. The comparison of tip diameters between the various conditions (Figs. 14a and 14b) shows that a correlation between tip diameter and heat flux does not exist. The shapes of the remaining fibers are similar for varying laser power. The fiber shapes are also similar at various locations within the crater (Fig. 14b). Essentially, in the sublimation regime, the shapes of the partially ablated fibers that still remain with the material are insensitive to the applied heat flux.

The needling of the fibers was further investigated to assess the mechanical effects of fiber deformation. The analysis is performed based on Eq. (1), which defines the momentum ratio, M_{ratio}, as the ratio between the momentum the fluid applies on the fiber and the momentum required to break the fiber [59]. The momentum ratio determines the likelihood that a fiber breaks under aerodynamic loads. The momentum ratio simplifies to the product of the surface area of the fiber, A_f, and the vertical center of mass location, $h_{f,\text{centroid}}$, divided by the moment of inertia of the fiber, I_{fiber}. Then, the ratio between M_{ratio} for the needled fibers and undeformed fibers can be computed to determine which fiber version is more likely to break, as shown in Eq. (2).

$$M_{\text{ratio}} = \frac{M_{\text{fluid}}}{M_{\text{break}}} = \frac{A_f h_{f,\text{centroid}}}{I_{\text{fiber}}} \tag{1}$$

$$M_{\text{undeformed/needle}} = \frac{M_{\text{ratio, undeformed}}}{M_{\text{ratio, needle}}} \tag{2}$$

A large number of needle fibers were cataloged from the SEM images for all three conditions. For each fiber, the cone angle and tip diameter were collected to define the needle fiber geometry. Then, M_{ratio} is computed for the needle fiber and the undeformed version of the fiber. The needle fiber geometry was defined by using a default base diameter of 6.6 µm, and computing the fiber length from the tip diameter and the cone angle. The undeformed fiber is defined as a cylinder with the same base diameter and length as the needle fiber. Finally, the ratio of M_{ratio} between the needle and undeformed fibers is computed for each cataloged fiber. The average ratio for each condition is shown in Table 3. Condition 1, the low power condition, presents the highest $M_{\text{undeformed/needle}}$ ratio among the three conditions, which indicates its undeformed fibers are more likely to break than its needled fibers, in comparison to conditions 2 and 3. However, the ratios across all three conditions are similar, corroborating the previous finding that the remaining fiber shapes are insensitive to the applied heat flux.

Table 3
Average ratio of M_{ratio} between needle and undeformed fibers for each condition.

	Condition 1	Condition 2	Condition 3
$M_{\text{undeformed/needle}}$	1.72	1.59	1.58

Overall, the analysis of the ablated materials indicates that a combination of material architectural features and heating mechanisms can lead to material evolution occurring through all differential ablation scenarios. Fig. 12a shows the fibers ablating at a faster or similar rate to the matrix, whereas Fig. 13a shows a region where the matrix ablates faster than the fibers. In regions where the fibers ablate faster than the matrix, the fibers recede into a thin wall of surrounding matrix. The fibers exhibit needling and thinning in regions where the matrix ablates faster than the fibers. In essence, the differential ablation scenarios can occur simultaneously under the same heating conditions. Samples under conditions 2 and 3 were also imaged and showed similar characteristics. These processes result from a combination of heating load and microstructural features of the C/C composites.

3.2.2. Evaluation of surface coating

Photographs of a C/C composite sample and graphite sample are shown in Figs. 15a and 15b, respectively. The surface of all samples had a metallic gray color propagating out from the site of laser impact. Micrographs of the gray surface obtained using SEM for two samples under condition 1 are shown in Fig. 15c and Fig. 15d. The images of the locations containing the gray surface are similar for the graphite and C/C composite samples despite differences in surface and internal material uniformity. The thickness of this surface coating is estimated to be around 100 µm. The non-ablated surface of both materials is shown in Fig. 1.

X-ray photoelectron spectroscopy (XPS) was used to determine the elemental composition and chemical state of the surface coating formed on the ablated samples. The results of the XPS scan provide insight into the formation of the surface coating and rule out possible causes such as reactions with unknown species in the vacuum chamber. The results of the XPS survey and carbon scans for graphite and C/C composite

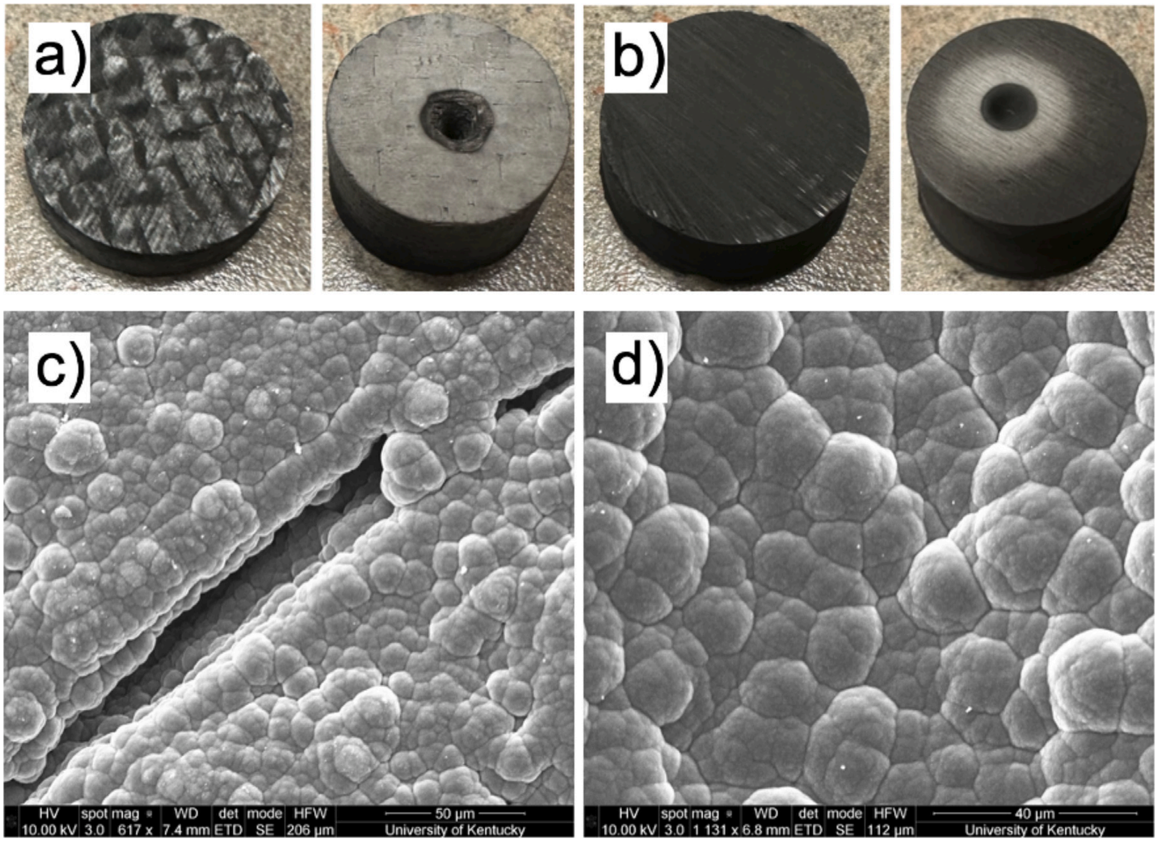

Fig. 15. Images of samples with gray surface coating and SEM micrographs of the gray coating for both samples. (a) Left: non-ablated C/C composite sample, right: ablated C/C composite sample, (b) left: non-ablated graphite sample, right: ablated graphite sample, (c) SEM micrograph of surface coating on C/C composite, (d) SEM micrograph of surface coating on graphite. XPS scans are only performed on the surface coating formed on the material.

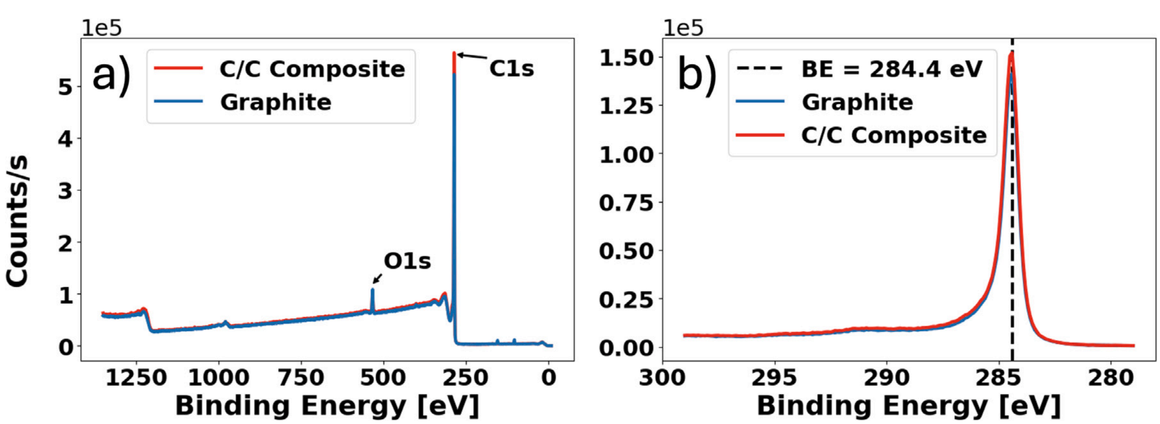

Fig. 16. Results of XPS scans. (a) Survey scan of graphite and C/C composite, (b) carbon scan of graphite and C/C composite.

are shown in Fig. 16. There are no notable differences between each scan location, so only the results from a single location are included. The survey scans shown in Fig. 16a indicate that the majority of the gray coating of the sample surface is carbon. Fig. 16b shows that the surface coating has a peak binding energy of 284.4 eV, consistent with the binding energy of sp^2 carbon [60].

The XPS survey results indicate that no unknown species in the chamber could have reacted with the carbon samples to form the surface coating. The dominant species in the surface coating is carbon. Specifically, the surface coating on graphite contained 92.9% carbon and 4.29% oxygen, whereas the coating on C/C composite contained 95.37% carbon and 3.38% oxygen. The remaining surface composition contained minuscule amounts (< 3%) of silicon, nitrogen, and sodium. The results of the high-resolution carbon scan indicate that the chemical state of the carbon is likely dominated by sp^2 carbon, as a result of the sublimation process that occurred during the experiment. We attribute the surface coating to the re-deposition of gaseous carbon species back to the surface. This re-deposition has been observed in previous studies that used microwave or inductively coupled plasma [61,62]. In both investigations, a low-density carbon preform called FiberForm was used that only contains the carbon phase. This observation has not been reported for sublimation nor observed on graphite or high-density C/C composites. However, it is not surprising that this re-deposition would occur during sublimation. Sublimation is a

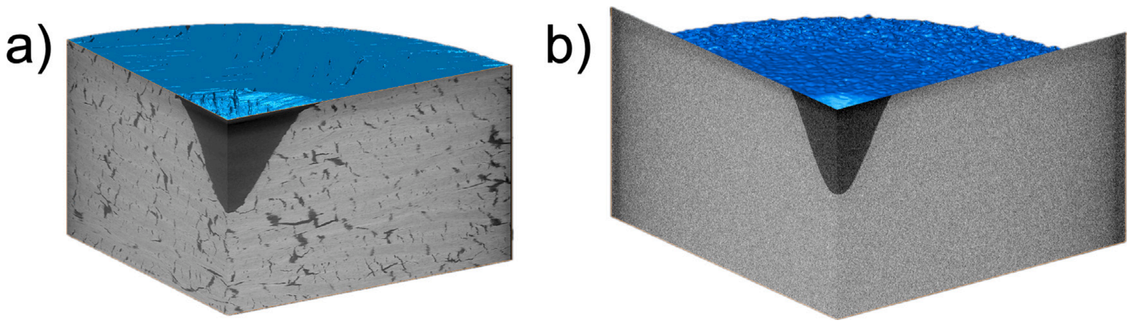

Fig. 17. XRCT tomographs showing internal material structure of (a) C/C composite and (b) graphite near the ablated region, indicating no sub-surface damage outside the crater.

phase-change process that is strongly dependent on temperature. As the laser is turned off and the sample cools, the gaseous carbon around the samples would re-deposit on the surface. Based on the image contrast of the SEM scans, the granules formed from the re-deposition have a lower density than the base material and are likely disordered carbon. The exact differences in densities cannot be ascertained from the contrast and will be the focus of future work.

The re-deposition will likely increase the thermal conductivity of the sample, leading to localized increase in heat load. The gray coating will also lower the surface emissivity of the material. Depending on the heating conditions, lower emissivity could be beneficial as the incoming radiative heat from the shock layer would be scattered away. However, the lower emissivity could also be detrimental because the emissions generated from the high temperatures within the material would get trapped, thereby increasing the thermal load inside the material. These interdependent processes will compete with each other if the materials are exposed to heat again. Thermal cycling of post-test composites and their subsequent material response will be the subject of future investigations. It must be emphasized here that the re-deposition of carbon could be an artifact of ground facilities and may not occur in real high-speed flight.

3.2.3. Crater morphology and mass loss through x-ray computed tomography

With the surface response of the samples analyzed through SEM and XPS, the internal material response was analyzed to further characterize the effects of the laser on graphite and C/C composite samples. Fig. 17 shows a tomograph of the ablated region and surrounding non-ablated material for both graphite and C/C composite. Fig. 17a shows the non-uniform internal material structure of the C/C composite, which was previously discussed in Section 2.1. In addition to the presence of the crater, the microstructure of the tested samples did not exhibit discernible differences from its original state, as shown in Fig. 17a. The effects of the internal material damage caused by the laser on graphite, shown in Fig. 17b, further support that the laser did not damage the sample outside the crater region. Fig. 17 shows no difference between the material adjacent to the crater and the non-ablated material further from the crater for the graphite sample. The uniform material structure of graphite described in Section 2.1, and shown on the surface in Fig. 1d, is consistent with the uniform material structure shown in Fig. 17b. The qualitative analysis provided through XRCT and SEM micrographs is further supported by quantitative information on the resulting depths of each crater.

The crater profiles obtained from XRCT are fit to a bivariate normal distribution using the curve-fit and least squares algorithms of SciPy [63]. The bivariate normal distribution equation is shown in Eq. (3), where A is the amplitude, and σ_x and σ_y are the spreads in the x and y directions. The crater profiles generated by the XRCT scans are loaded into the program as a stereolithography (STL) mesh. The program uses the curve-fitting algorithm and the non-linear least squares algorithm from the optimization module of SciPy to fit the parameters of Eq. (3) to the XRCT mesh. The best fit between the curve fitting method and the non-linear least squares method is chosen based on the similarity between the generated and the XRCT meshes. Fig. 18 shows the process used to create an STL mesh from the bivariate normal fit of a crater profile. Fig. 18a shows a C/C composite sample after the experiment, Fig. 18b shows the XRCT reconstruction of the sample with the extracted crater profile, and Fig. 18c shows the resulting bivariate normal distribution fit overlayed with the XRCT scan. Table 4 shows a summary of the fitted parameters for each condition, with the mean of the parameters over the various samples under the same condition. A complete list of parameters is shown in the Supporting Information Table S4.

Table 4
The average bivariate distribution parameters for each sample and condition. The full list of parameters is found in Table S4 in the Supporting Information.

	Graphite			C/C composite		
	A [mm]	σ_x [mm]	σ_y [mm]	A [mm]	σ_x [mm]	σ_y [mm]
Condition 1	2.63	0.92	0.91	2.61	0.92	0.93
Condition 2	8.24	0.97	0.98	7.29	1.02	1.03
Condition 3	12.06	0.64	0.70	9.45	0.62	0.65

$$z(x, y) = A \exp\left(-\frac{x^2}{2\sigma_x^2} - \frac{y^2}{2\sigma_y^2}\right) \tag{3}$$

The amplitude A from Table 4 shows the average depths of each material and condition. Considering that the samples under condition 3 were exposed for different amounts of time, the recession shown in Table 4 for both materials under condition 3 is from a single sample each, which was exposed for 25 s. The crater recession increases exponentially as the set laser power of each condition increases. The Knudsen–Langmuir equation has been previously used to predict the recession rate of carbon caused by sublimation in high-temperature, low-pressure conditions [17,64]. The Knudsen–Langmuir equation predicts an exponential increase in recession rate as a function of temperature, which is consistent with our observations.

For condition 1, the graphite and C/C composite have similar average depths represented by the amplitude parameter for the bivariate Gaussian fit. The C/C composite has a lower amplitude for conditions 2 and 3 than graphite, indicating a lower average depth. In general, graphite produces a deeper crater when exposed to the laser, whereas C/C composite produces a shallower crater. The difference in crater shape can be attributed to the difference in conductivity. As observed in the material response simulation of condition 1 (Fig. 8), the heat flux concentrates within the front face of the C/C composite, creating a shallow high-temperature region where recession occurs. In contrast, the heat flux can penetrate the graphite material further than it does in the C/C composite, creating a deep high-temperature region where recession occurs. Similar behavior was also observed when simulating

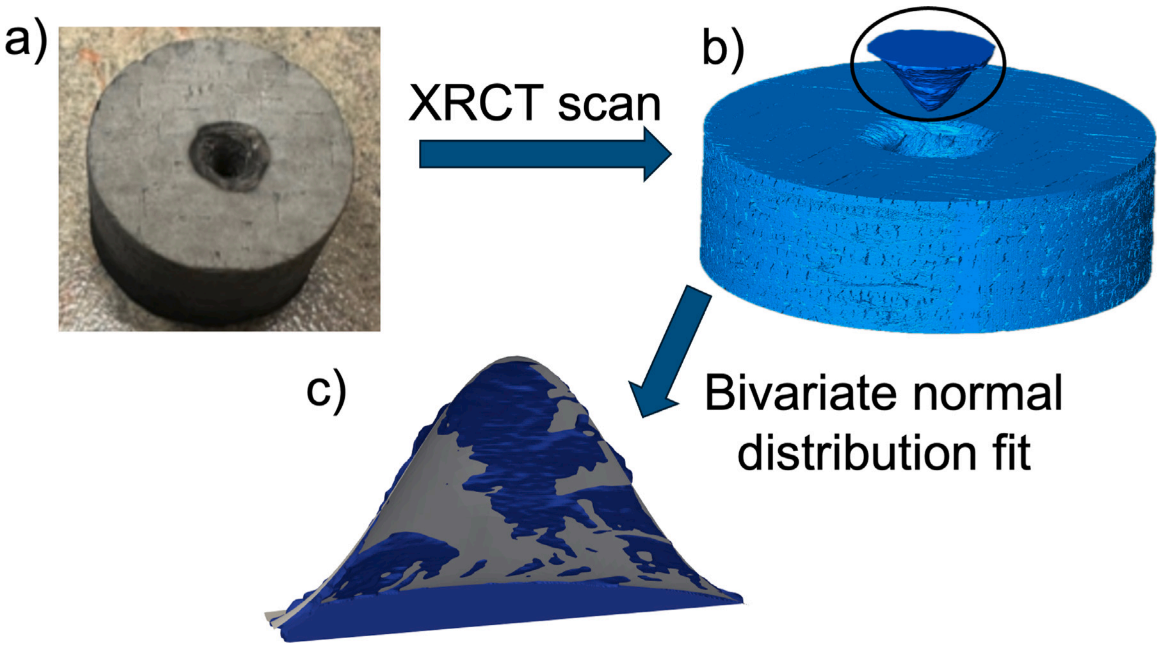

Fig. 18. Process to generate the bivariate normal fit for a sample. (a) An image of an ablated C/C composite sample, (b) a mesh of the sample and crater profile generated using XRCT, (c) the crater mesh (blue) overlayed with a new mesh generated from the bivariate normal distribution fit (gray).

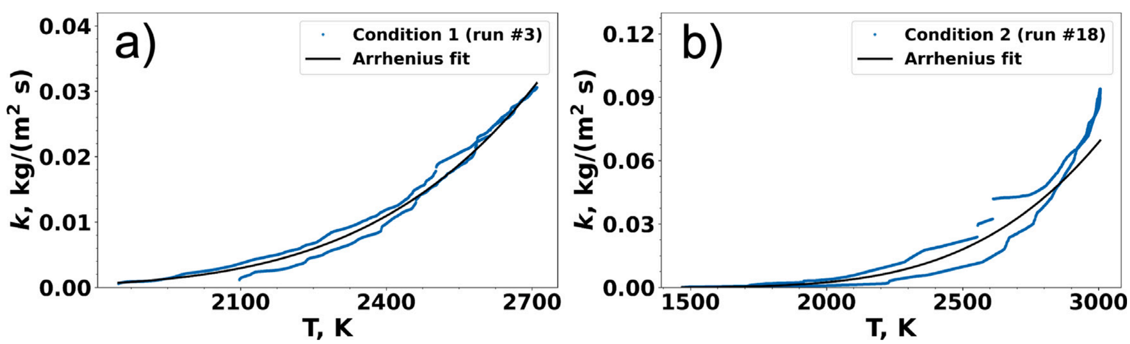

Fig. 19. Comparison of Arrhenius equation fit to XRCT scan slices for each condition. (a) Condition 1, run #3, and (b) Condition 2, run #18.

conditions 2 and 3, as shown in Figs. S9 and S10 of the Supporting Information. The heat flux concentration on the front of the C/C composite can be explained by the composite having a larger in-plane than through-plane conductivity. In contrast, graphite has isotropic conductivity, which allows the heat to diffuse equally in all directions, penetrating the sample deeper than it does in the C/C composite.

Finally, the XRCT scans were also used to estimate the rate of carbon sublimation using the Arrhenius equation (Eq. (4)). First, 2D slices across the center of the crater were extracted from the XRCT scans. From the slices, the rate k was estimated by dividing the depth of recession along each slice by the total experimental duration and multiplying by the carbon density. The surface temperature for each slice was estimated by performing a surface energy balance using the laser heat flux, the conduction into the material, and the re-radiation heat flux. With the temperature distribution, the Arrhenius equation was fit to match the rate k distribution for each slice, the optimal pre-exponential factor A and activation energy E_a were obtained from the fit.

$$k = Ae^{\left(\frac{-E_a}{RT}\right)} \tag{4}$$

The temperature is obtained by performing an energy balance at the front surface of the sample. The surface receives the laser heat flux, emits re-radiation, and absorbs the energy via conduction. The energy balance equation is shown in Eq. (5). The first term represents the laser heat flux, where P_o is the laser power, A is the temperature-dependent laser absorption, η is the laser efficiency, r_o is the laser beam width, and r_f is the distance from the laser beam to the target face [36]. A full estimation of the laser heat flux is shown in Fig. S1 in the Supporting Information. The second term is the conduction into the material, where k_s is the thermal conductivity of the material, T is the surface temperature, T_0 is the initial surface temperature set to 300 K, α_s is the diffusivity of the material, and t is the experiment duration. The final term is the re-radiation, where M_{rad} is a multiplier term added to account for the additional surfaces emitting re-radiation, ε is the emissivity of the material, and σ is the Stefan–Boltzmann constant. Eq. (5) is solved numerically for the temperature.

$$\frac{2P_oA\eta}{\pi r_o^2}e^{\left(\frac{-2r_f^2}{r_o^2}\right)} = \frac{k_s(T-T_0)}{2\sqrt{\alpha_s t/\pi}} + M_{\text{rad}}\varepsilon\sigma(T^4 - T_0^4) \tag{5}$$

The best fit for the pre-exponential factor, A, and activation energy, E_a, was computed for each XRCT scan. The most accurate fits for conditions 1 and 2 are shown in Fig. 19. A general best fit was obtained by averaging the most accurate results from conditions 1 and 2. The pre-exponential factor was determined to be 81.12 kg/(m^2 s), and the activation energy was 175.14 kJ/mol.

4. Conclusions

Sublimation experiments on graphite and C/C composite have been conducted using a continuous wave laser in a low-pressure, inert atmosphere. Multiple samples of each material were ablated under laser powers of 900 W, 1350 W, and 1800 W. Temperature measurements showed that the graphite samples had a higher average back face temperature than the C/C composite samples for each condition.

The C/C composite has a larger in-plane conductivity as a result of the non-uniform internal material structure. Material response simulations showed that the in-plane conductivity caused the heat to concentrate on the impinging surface and diffuse within the plane, which led to larger cooling via radiation and a lower average back face temperature than graphite. Crater profile analysis using X-ray computed tomography showed that the crater depth produced by the laser was less for C/C composite samples than graphite, indicating the severity of damage was less for C/C composite than graphite. When graphite and C/C composite samples are exposed to a high power laser, C/C composite samples tend to form shallower craters while graphite samples form deeper craters.

SEM micrographs of C/C composite microstructures show evidence that the matrix ablates slower than the fibers in some areas, which has previously been observed under sublimation conditions. Additional SEM micrographs show regions of the ablated C/C composite with a cluster of fibers that lack matrix, either from a lack of matrix in the non-ablated material or the contradictory scenario, where the matrix ablates faster than the fibers under sublimation. The fibers that appear in these regions have needle-shaped faceted points, unlike those that occur under oxidation, which are smooth and conical. The needling and thinning of fibers has also been observed, which is frequently seen under oxidation or under both oxidation and sublimation conditions, but has not been reported under isolated sublimation conditions. Surface images of regions surrounding the crater show the development of a surface coating on both materials. The classification of the surface coating using XPS determined that the coating was sp^2 carbon, which was attributed to the re-deposition of sublimation gases on the surface, a process called re-deposition.

CRediT authorship contribution statement

Frederick N. Shireman: Writing – review & editing, Writing – original draft, Visualization, Validation, Investigation, Formal analysis, Data curation. **Bruno Tacchi:** Writing – review & editing, Writing – original draft, Visualization, Validation, Investigation, Formal analysis, Data curation. **Kate B. Rhoads:** Writing – review & editing, Investigation, Formal analysis, Data curation. **Tyler C. Haag:** Writing – review & editing, Investigation. **Ryan J. Thorne:** Writing – review & editing, Visualization, Investigation. **Boris S. Leonov:** Writing – review & editing, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation. **Alexandre Martin:** Writing – review & editing, Project administration, Investigation, Funding acquisition, Data curation, Conceptualization. **Savio J. Poovathingal:** Writing – review & editing, Writing – original draft, Software, Resources, Project administration, Investigation, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.ijheatmasstransfer.2026.128878.

References

[1] E. Zadorozny, Space shuttle carbon-carbon composite hot structure, in: 24th Structures, Structural Dynamics and Materials Conference, American Institute of Aeronautics and Astronautics, Lake Tahoe, Nevada, 1983, p. 913, http://dx.doi.org/10.2514/6.1983-913.

[2] P. Garee, S. Sundaram, et al., Carbon fiber reinforced plastic composite structural analysis on a reentry capsule heat shield, Mater. Today: Proc. 4 (8) (2017) 8200–8207, http://dx.doi.org/10.1016/j.matpr.2017.07.162.

[3] K.K. Kuo, S. Keswani, A comprehensive theoretical model for carbon-carbon composite nozzle recession, Combust. Sci. Technol. 42 (3–4) (1985) 145–164, http://dx.doi.org/10.1080/00102208508960374.

[4] V. Borie, J. Brulard, G. Lengelle, Aerothermochemical analysis of carbon-carbon nozzle regression in solid-propellant rocket motors, J. Propuls. Power 5 (6) (1989) 665–673, http://dx.doi.org/10.2514/3.23204.

[5] D. Boury, L. Filipuzzi, SEPCARB materials for solid rocket booster nozzle components, in: 37th Joint Propulsion Conference and Exhibit, American Institute of Aeronautics and Astronautics, Salt Lake City, Utah, 2001, p. 3438, http://dx.doi.org/10.2514/6.2001-3438.

[6] E. Fitzer, L.M. Manocha, Carbon Reinforcements and Carbon/Carbon Composites, Springer Science & Business Media, Berlin, Germany, 1998.

[7] X. Li, W. Strieder, Emissivity of high-temperature fiber composites, Ind. Eng. Chem. Res. 48 (4) (2009) 2236–2244, http://dx.doi.org/10.1021/ie8008583.

[8] S. Pestchanyi, V. Safronov, I. Landman, Estimation of carbon fibre composites as ITER divertor armour, J. Nucl. Mater. 329 (2004) 697–701, http://dx.doi.org/10.1016/j.jnucmat.2004.04.189.

[9] D. Duffy, Fusion power materials, Adv. Mater. Process. 168 (7) (2010).

[10] Q. Zhang, D. Lan, S. Deng, J. Gu, Y. Wang, J. Ren, G. Wu, Z. Jia, Constructing multiple heterogeneous interfaces in one-dimensional carbon fiber materials for superior electromagnetic wave absorption, Carbon 226 (2024) 119233, http://dx.doi.org/10.1016/j.carbon.2024.119233.

[11] C.-M. Fan, Q. Zhang, Y. Wang, Y. Jin, Q. He, Constructing core-shell SiC@ c nanofiber network structure to enhance conductive loss for efficient microwave absorption, Inorg. Chem. Front. (2025) http://dx.doi.org/10.1039/D5QI01376C.

[12] Y. Lei, Q. He, Y. Wang, C. Fan, X. Yin, C. Wang, L. Liu, Sustainable cofe-doped biomass carbon aerogels for ultra-broadband electromagnetic absorption and thermal management, ACS Sustain. Chem. Eng. (2025) http://dx.doi.org/10.1021/acssuschemeng.5c07718.

[13] D. Wu, D. Lan, S. Zhang, Q. He, X. Zhou, Y. Wang, Lightweight dielectric-magnetic synergistic necklace-shaped co @ NCP/carbon nanofiber composites for enhanced electromagnetic wave absorption, Mater. Today Nano 28 (2024) 100520, http://dx.doi.org/10.1016/j.mtnano.2024.100520.

[14] G. Duffa, Ablative Thermal Protection Systems Modeling, American Institute of Aeronautics and Astronautics, Inc., 2013.

[15] W.H. Bowman, R.M. Lawrence, Ablative materials for high-temperature thermal protection of space vehicles, J. Chem. Educ. 48 (10) (1971) 690–691, http://dx.doi.org/10.1021/ed048p690.

[16] G.V. Candler, Rate effects in hypersonic flows, Annu. Rev. Fluid Mech. 51 (1) (2019) 379–402, http://dx.doi.org/10.1146/annurev-fluid-010518-040258.

[17] C. Levet, J. Lachaud, V. Ducamp, R. Memes, J. Couzi, J. Mathiaud, A.P. Gillard, P. Weisbecker, G.L. Vignoles, High-flux sublimation of a 3D carbon/carbon composite: Surface roughness patterns, Carbon 173 (2021) 817–831, http://dx.doi.org/10.1016/j.carbon.2020.11.023.

[18] L. Geng, X. Liu, Q. Fu, S. Cheng, H. Li, Laser ablative behavior of C/C modified by si reactive infiltration, Carbon 168 (2020) 650–658, http://dx.doi.org/10.1016/j.carbon.2020.06.079.

[19] L. Geng, S. Cheng, Q. Fu, H. Li, Laser ablation behavior and mechanism of carbon/carbon composites, Fuhe Cailiao Xuebao/Acta Mater. Compos. Sin. 39 (9) (2022) 4337–4343, http://dx.doi.org/10.13801/j.cnki.fhclxb.20220825.002.

[20] G.L. Vignoles, J. Lachaud, Y. Aspa, J.-M. Goyhénèche, Ablation of carbon-based materials: Multiscale roughness modelling, Compos. Sci. Technol. 69 (9) (2009) 1470–1477, http://dx.doi.org/10.1016/j.compscitech.2008.09.019.

[21] E.C. Stern, Microscale Modeling of Porous Thermal Protection System Materials (Ph.D. thesis), University of Minnesota, 2015.

[22] F. Panerai, N.N. Mansour, J. Lachaud, A. Martin, Numerical and experimental study of carbon fiber oxidation, in: 52nd Aerospace Sciences Meeting, American Institute of Aeronautics and Astronautics, National Harbor, Maryland, 2014, p. 1208, http://dx.doi.org/10.2514/6.2014-1208.

[23] F. Rodriguez-Reinoso, P. Thrower, Microscopic studies of oxidized highly oriented pyrolytic graphites, Carbon 12 (3) (1974) 269–279, http://dx.doi.org/10.1016/0008-6223(74)90069-4.

[24] C. Wong, R.T. Yang, B.L. Halpern, The mode of attack of oxygen atoms on the basal plane of graphite, J. Chem. Phys. 78 (6) (1983) 3325–3328, http://dx.doi.org/10.1063/1.445198.

[25] X. Chu, L. Schmidt, Gasification of graphite studied by scanning tunneling microscopy, Carbon 29 (8) (1991) 1251–1255, http://dx.doi.org/10.1016/0008-6223(91)90044-J.

[26] L. Zhang, D.A. Pejaković, B. Geng, J. Marschall, Surface modification of highly oriented pyrolytic graphite by reaction with atomic nitrogen at high temperatures, Appl. Surf. Sci. 257 (13) (2011) 5647–5656, http://dx.doi.org/10.1016/j.apsusc.2011.01.066.

[27] J.E. Sheehan, K. Buesking, B. Sullivan, Carbon-carbon composites, Annu. Rev. Mater. Sci. 24 (1) (1994) 19–44, http://dx.doi.org/10.1146/annurev.ms.24.080194.000315.

[28] J. Lachaud, N. Bertrand, G.L. Vignoles, G. Bourget, F. Rebillat, P. Weisbecker, A theoretical/experimental approach to the intrinsic oxidation reactivities of C/C composites and of their components, Carbon 45 (14) (2007) 2768–2776, http://dx.doi.org/10.1016/j.carbon.2007.09.034.

[29] J. Lachaud, Y. Aspa, G. Vignoles, J.-M. Goyhéneche, 3D modeling of thermochemical ablation in carbon-based materials: effect of anisotropy on surface roughness onset, in: Proceedings of the Tenth International Symposium on Materials in a Space Environment, vol. 616, European Space Agency, Noordwijk, The Netherlands, 2006, p. 10.

[30] S. Poovathingal, T.E. Schwartzentruber, V.J. Murray, T.K. Minton, Molecular simulation of carbon ablation using beam experiments and resolved microstructure, AIAA J. 54 (3) (2016) 999–1010, http://dx.doi.org/10.2514/1.J054562.

[31] D. Liu, H. Chen, L.A. Chacon, V.M. Ramu, S.J. Poovathingal, Micro-CT image-based computation of effective thermal and mechanical properties of fibrous porous materials, Compos. Part B: Eng. 281 (2024) 111502, http://dx.doi.org/10.1016/j.compositesb.2024.111502.

[32] J. Lachaud, Y. Aspa, G.L. Vignoles, Analytical modeling of the steady state ablation of a 3D c/c composite, Int. J. Heat Mass Transfer 51 (9–10) (2008) 2614–2627, http://dx.doi.org/10.1016/j.ijheatmasstransfer.2008.01.008.

[33] J. Keenan, G. Candler, Simulation of graphite sublimation and oxidation under re-entry conditions, in: 6th Joint Thermophysics and Heat Transfer Conference, American Institute of Aeronautics and Astronautics, Colorado Springs, Colorado, 1994, p. 2083, http://dx.doi.org/10.2514/6.2012-724.

[34] S. Poovathingal, T.E. Schwartzentruber, V.J. Murray, T.K. Minton, G.V. Candler, Finite-rate oxidation model for carbon surfaces from molecular beam experiments, AIAA J. 55 (5) (2017) 1644–1658, http://dx.doi.org/10.2514/1.J055371.

[35] N. Mullenix, A. Povitsky, Hypersonic ablation of graphite thermal protection systems with surface defects, J. Spacecr. Rockets 53 (5) (2016) 912–929, http://dx.doi.org/10.2514/1.A33380.

[36] A.C. Thombs, B.S. Leonov, J. Meyers, R.B. Miles, J. Grinstead, Finite volume modeling of graphite heat transfer and sublimation during high-power laser heating, in: AIAA Aviation Forum and ASCEND 2024, American Institute of Aeronautics and Astronautics, Las Vegas, Nevada, 2024, p. 3985, http://dx.doi.org/10.2514/6.2024-3985.

[37] I. Entegris, Graphite properties and characteristics, 2020, https://poco.entegris.com/content/dam/poco/resources/reference-materials/brochures/brochure-graphite-properties-and-characteristics-11043.pdf; (Accessed 27 June 2025).

[38] Americarb, Americarb, 2025, URL https://www.americarb.com, (Date Accessed 11 August 2025).

[39] P. Cloetens, M. Pateyron-Salomé, J.Y. Buffière, G. Peix, J. Baruchel, F. Peyrin, M. Schlenker, Observation of microstructure and damage in materials by phase sensitive radiography and tomography, J. Appl. Phys. 81 (9) (1997) 5878–5886, http://dx.doi.org/10.1063/1.364374.

[40] H. Weng, A. Martin, Multidimensional modeling of pyrolysis gas transport inside charring ablative materials, J. Thermophys. Heat Transfer 28 (4) (2014) 583–597, http://dx.doi.org/10.2514/1.T4434.

[41] C. Mortensen, X. Zhong, High-order shock-fitting method for hypersonic flow with graphite ablation and boundary layer stability, in: 42nd AIAA Fluid Dynamics Conference and Exhibit, American Institute of Aeronautics and Astronautics, New Orleans, LA, 2012, p. 3150, http://dx.doi.org/10.2514/6.2012-3150.

[42] T. Dolton, H. Goldstein, R. Maurer, Thermodynamic performance of carbon in hyperthermal environments, in: 3rd Thermophysics Conference, American Institute of Aeronautics and Astronautics, Los Angeles, CA, 1968, p. 754, http://dx.doi.org/10.2514/6.1968-754.

[43] J. Brooks, G. Taylor, in: P. walker (Ed.), Chemistry and Physics of Carbon, vol. 4, Jr. Marcel Dekker Inc., NY, 1968, p. 243.

[44] B.D. Tacchi, F. Shireman, A.H. Yassin, K.B. Rhoads, A. Martin, S.J. Poovathingal, B.S. Leonov, T.C. Haag, R.B. Miles, Modeling the ablation of carbon-based materials through continuous-wave lasers, in: AIAA SCITECH 2026 Forum, American Institute of Aeronautics and Astronautics, Orlando, FL, 2026, p. 0232, http://dx.doi.org/10.2514/6.2026-0232.

[45] J. Lachaud, Y. Aspa, G.L. Vignoles, Analytical modeling of the transient ablation of a 3D C/C composite, Int. J. Heat Mass Transfer 115 (2017) 1150–1165, http://dx.doi.org/10.1016/j.ijheatmasstransfer.2017.06.130.

[46] G.L. Vignoles, J. Lachaud, Y. Aspa, Environmental effects: Ablation of C/C materials—Surface dynamics and effective reactivity, in: Ceramic Matrix Composites, John Wiley & Sons, Ltd, 2014, pp. 353–388, http://dx.doi.org/10.1002/9781118832998.ch12.

[47] S. Qingliang, W. Tiyuan, S. Qiang, Y. Fang, L. Hejun, M.W. Fu, Unraveling of the laser drilling of carbon/carbon composites: Ablation mechanisms, shape evolution, and damage evaluation, Int. J. Mach. Tools Manuf. 184 (2023) 103978, http://dx.doi.org/10.1016/j.ijmachtools.2022.103978.

[48] J. Zhang, R. Bi, S. Jiang, Z. Wen, C. Luo, J. Yao, G. Liu, C. Chen, M. Wang, Laser ablation mechanism and performance of carbon fiber-reinforced poly aryl ether ketone (Paek) composites, Polymers 14 (13) (2022) http://dx.doi.org/10.3390/polym14132676.

[49] D. Cho, B. Il Yoon, Microstructural interpretation of the effect of various matrices on the ablation properties of carbon-fiber-reinforced composites, Compos. Sci. Technol. 61 (2) (2001) 271–280, http://dx.doi.org/10.1016/S0266-3538(00)00212-8.

[50] B.A. Newcomb, Processing, structure, and properties of carbon fibers, Compos. Part A: Appl. Sci. Manuf. 91 (2016) 262–282, http://dx.doi.org/10.1016/j.compositesa.2016.10.018.

[51] X. Huang, Fabrication and properties of carbon fibers, Materials 2 (4) (2009) 2369–2403, http://dx.doi.org/10.3390/ma2042369.

[52] K. Chawla, Carbon fibers, in: Fibrous Materials, Cambridge University Press, 2016, pp. 230–253.

[53] S.-S. Tzeng, Y.-G. Chr, Evolution of microstructure and properties of phenolic resin-based carbon/carbon composites during pyrolysis, Mater. Chem. Phys. 73 (2–3) (2002) 162–169, http://dx.doi.org/10.1016/S0254-0584(01)00358-3.

[54] T.K. Schneck, B. Brück, M. Schulz, J.M. Spörl, F. Hermanutz, B. Clauß, W.M. Mueller, B. Heidenreich, D. Koch, S. Horn, et al., Carbon fiber surface modification for tailored fiber-matrix adhesion in the manufacture of C/C-SiC composites, Compos. Part A: Appl. Sci. Manuf. 120 (2019) 64–72, http://dx.doi.org/10.1016/j.compositesa.2019.02.021.

[55] S. Sharma, C. Shyam Kumar, J.G. Korvink, C. Kübel, Evolution of glassy carbon microstructure: In situ transmission electron microscopy of the pyrolysis process, Sci. Rep. 8 (1) (2018) 16282, http://dx.doi.org/10.1038/s41598-018-34644-9.

[56] T.J. Cochell, R.R. Unocic, J. Graña-Otero, A. Martin, Nanoscale oxidation behavior of carbon fibers revealed with in situ gas cell STEM, Scr. Mater. 199 (2021) 113820, http://dx.doi.org/10.1016/j.scriptamat.2021.113820.

[57] T.J. Cochell, R.R. Unocic, B.K. Bessire, S.J. Poovathingal, A. Martin, In-situ nanoscale ablation, in: AIAA AVIATION FORUM and ASCEND 2024, 2024, p. 3729, http://dx.doi.org/10.2514/6.2024-3729.

[58] H. Diaf, A. Pereira, P. Melinon, N. Blanchard, F. Bourquard, F. Garrelie, C. Donnet, M. Vondráček, Revisiting thin film of glassy carbon, Phys. Rev. Mater. 4 (6) (2020) 066002, http://dx.doi.org/10.1103/PhysRevMaterials.4.066002.

[59] M. Ziering, V. DiCristina, Thermomechanical erosion of ablative plastic composites, in: 7th Thermophysics Conference, American Institute of Aeronautics and Astronautics, San Diego, TX, 1972, p. 299, http://dx.doi.org/10.2514/6.1972-299.

[60] B. Lesiak, L. Kövér, J. Tóth, J. Zemek, P. Jiricek, A. Kromka, N. Rangam, C sp^2/sp^3 hybridisations in carbon nanomaterials - XPS and (x)AES study, Appl. Surf. Sci. 452 (2018) 223–231, http://dx.doi.org/10.1016/j.apsusc.2018.04.269.

[61] K.B. Rhoads, K.J. Price, I.M. Wadsworth, S. Loehle, S.J. Poovathingal, A. Martin, Evidence of spallation due to volumetric ablation, in: AIAA SCITECH 2025 Forum, American Institute of Aeronautics and Astronautics, Orlando, FL, 2025, p. 1425, http://dx.doi.org/10.2514/6.2025-1425.

[62] B.M. Ringel, H.J. Boesch, S. Oruganti, L. Villafañe, F. Panerai, Unsteady spallation of low-density carbon fiber ablators, Carbon 248 (2025) 121113, http://dx.doi.org/10.1016/j.carbon.2025.121113.

[63] P. Virtanen, R. Gommers, T.E. Oliphant, M. Haberland, T. Reddy, D. Cournapeau, E. Burovski, P. Peterson, W. Weckesser, J. Bright, et al., SciPy 1.0: fundamental algorithms for scientific computing in python, Nature Methods 17 (3) (2020) 261–272, http://dx.doi.org/10.1038/s41592-019-0686-2.

[64] A.L. Zibitsker, J.A. McQuaid, E.C. Stern, G.E. Palmer, B.J. Libben, C. Brehm, A. Martin, Finite-rate and equilibrium study of graphite ablation under arc-jet conditions, Comput. & Fluids 267 (2023) 106069, http://dx.doi.org/10.1016/j.compfluid.2023.106069.